



香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

ELEC 2100 Signals and Systems

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Outline Chapter 1 Basic concepts of signals and systems



THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

1 Basic concepts

2 Description and classification of signals

3 Continuous-time signals

4 Unit step signal and unit impulse signal

5 Decomposition of signals

6 Discrete-time signals

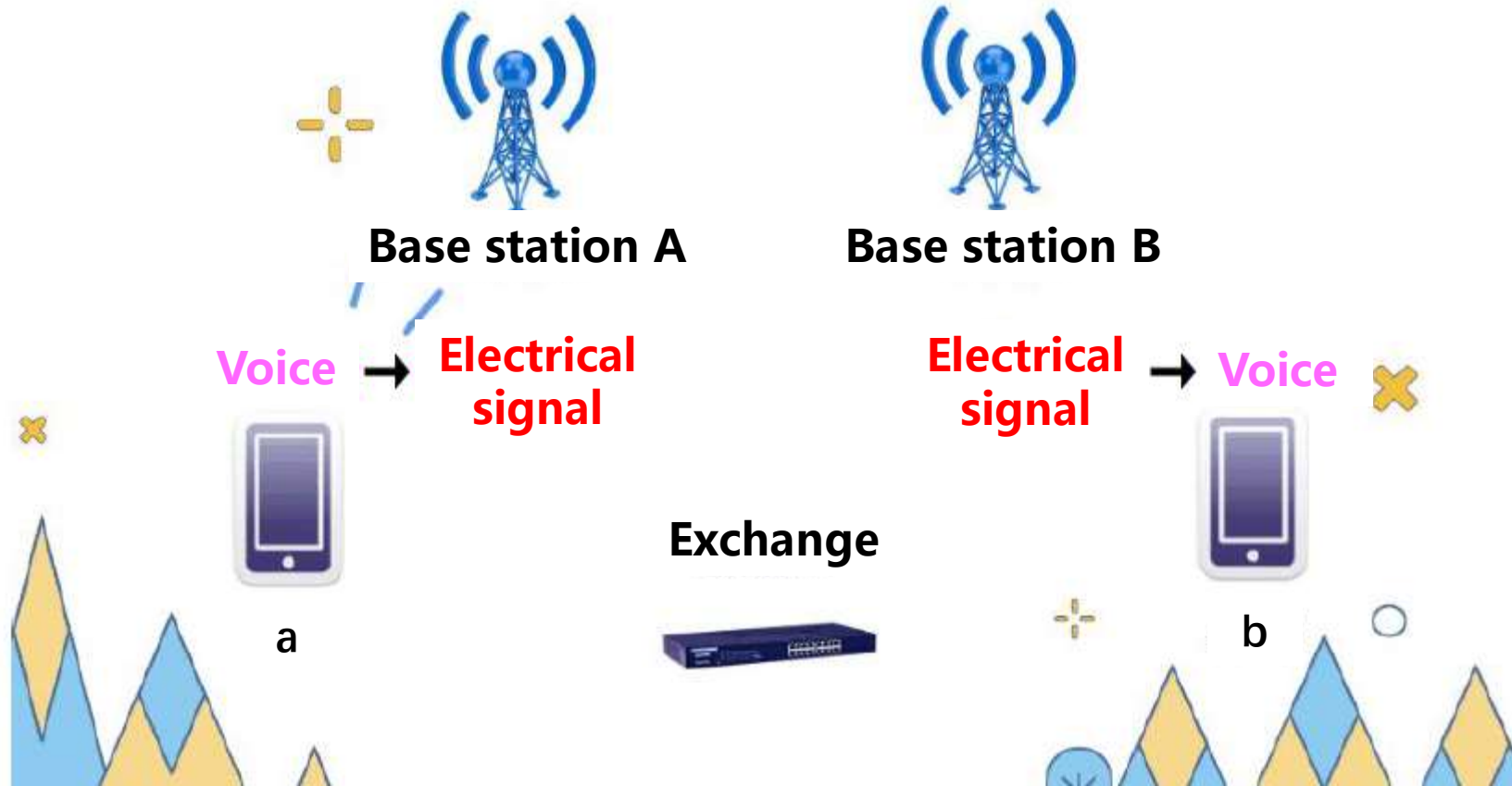
7 System

Main Content

1. Basic concepts of signals and systems
2. Theory of Signals and Systems
3. Learning method
4. Reference book

1. Basic concepts of signals and systems

1. Basic concepts of signals and systems



1. Basic concepts of signals and systems

(1) Signal

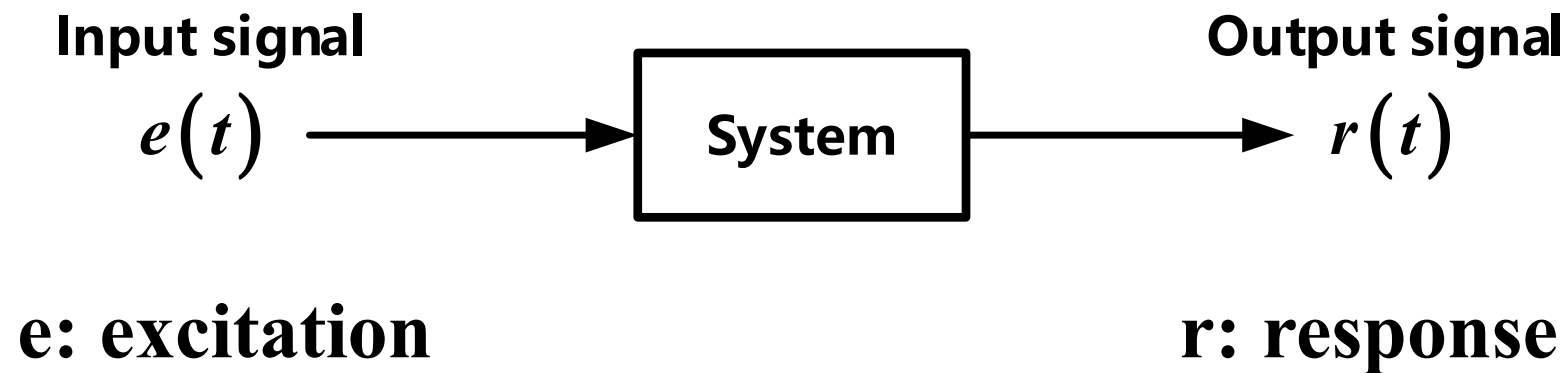
- **Message:** In communication systems, voice, text, images, and data are collectively referred to as messages
- **Signal: Refers to the representation and transmission medium of a message**
- A signal serves as both the representation and transmission medium of a message, while the message constitutes the actual content being conveyed by the signal

1. Basic concepts of signals and systems

(2) System

System

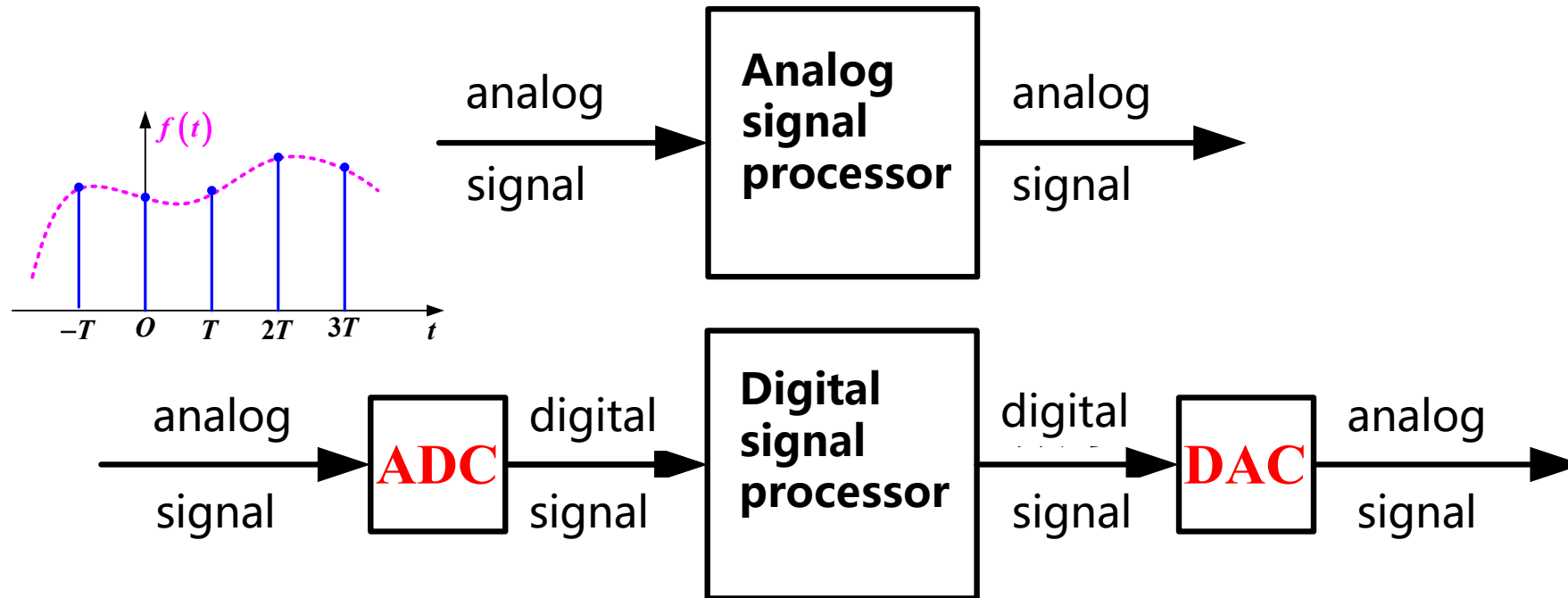
An integrated entity composed of multiple interacting and interdependent components, possessing stable functionality



1. Basic concepts of signals and systems

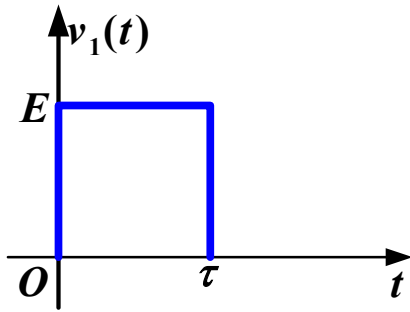
(2) System

The basic function of a system: To process and transform input signals into the desired output signals

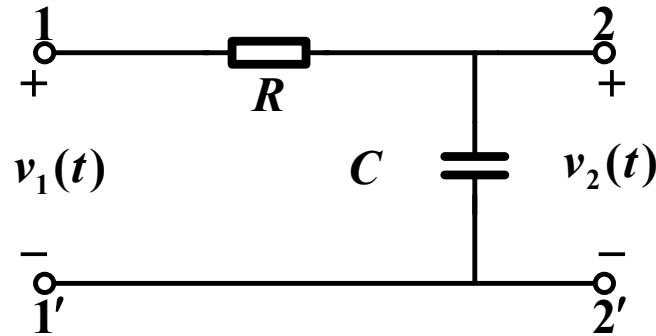


1. Basic concepts of signals and systems

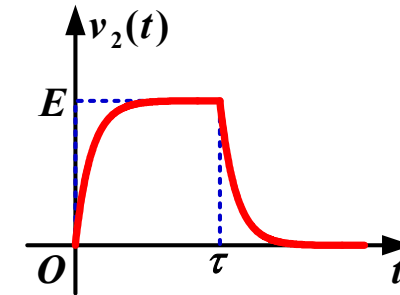
(2) System



Input signal



System



Output signal

- Will the input signal be distorted after transmission through the system?
- How does the system affect the transmission of the input signal?
- Does the system process the signal to achieve the desired effect?

1. Basic concepts of signals and systems

2. Theory of Signals and Systems

Signal Theory

Signal Analysis:
Studies the
description and
properties of
signals

Signal Transmission

Signal Processing

System Theory

System Analysis:
For a given system,
examines the
output generated
in response to
input signals

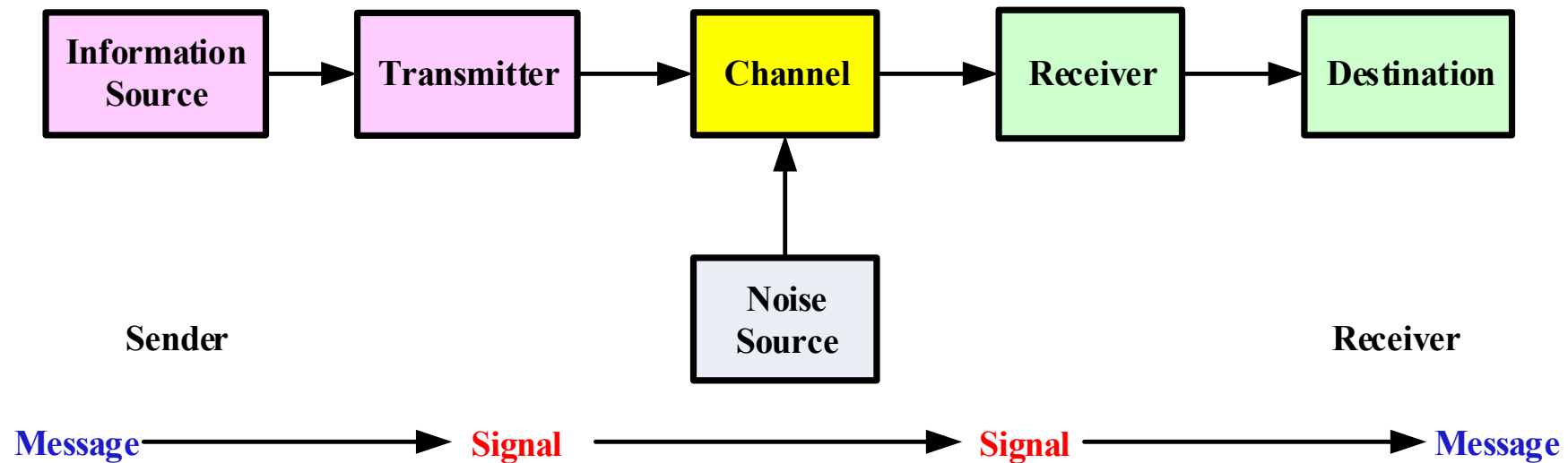
System Synthesis:
Designs
(synthesizes)
systems according
to specified
requirements

1. Basic concepts of signals and systems

(1) Communication system

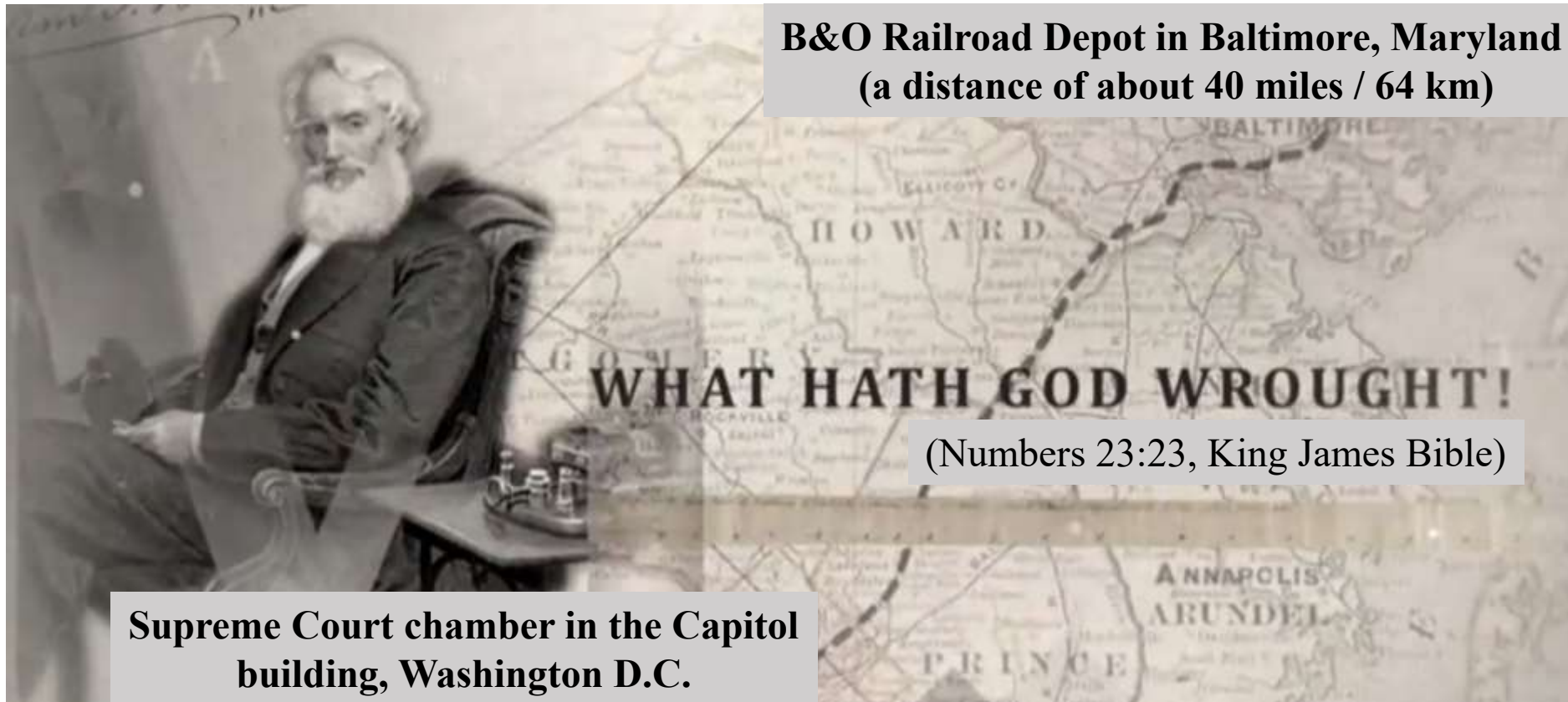
The purpose of communication is to achieve message transmission

Communication System: A complete set of technical equipment (including transmission channels) installed for the purpose of message delivery



1. Basic concepts of signals and systems

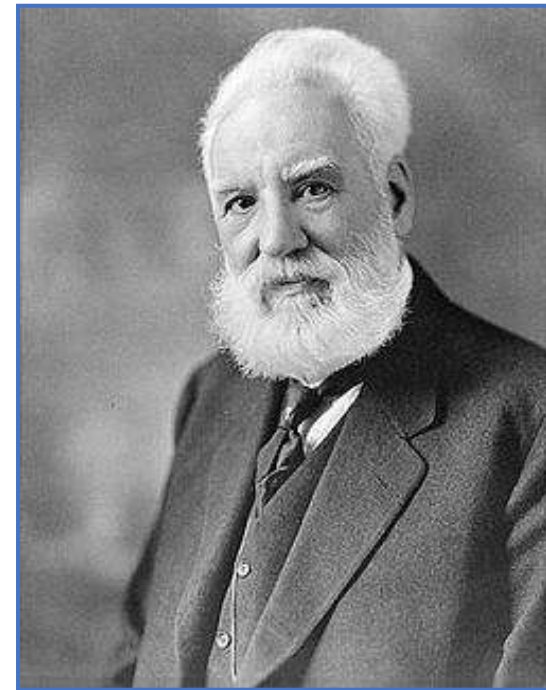
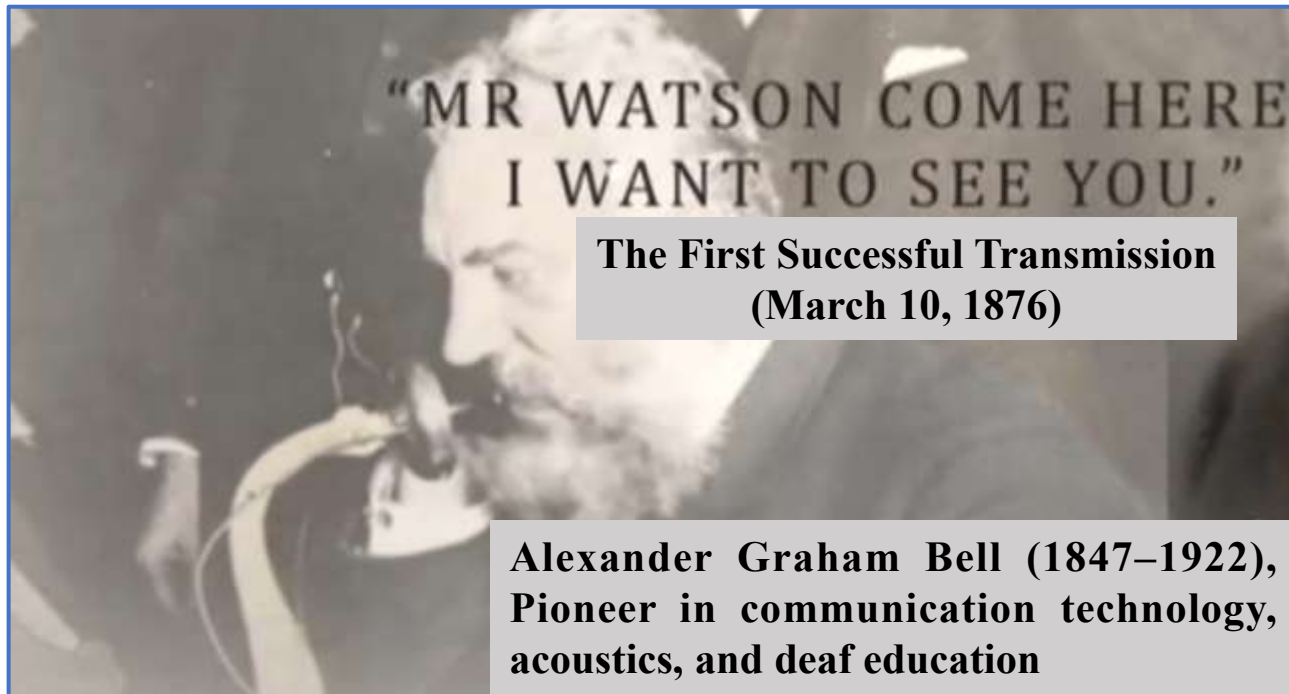
(1) Communication system: uses electrical signals to transmit messages



In 1844, Morse successfully transmitted the first telegraph message in history

1. Basic concepts of signals and systems

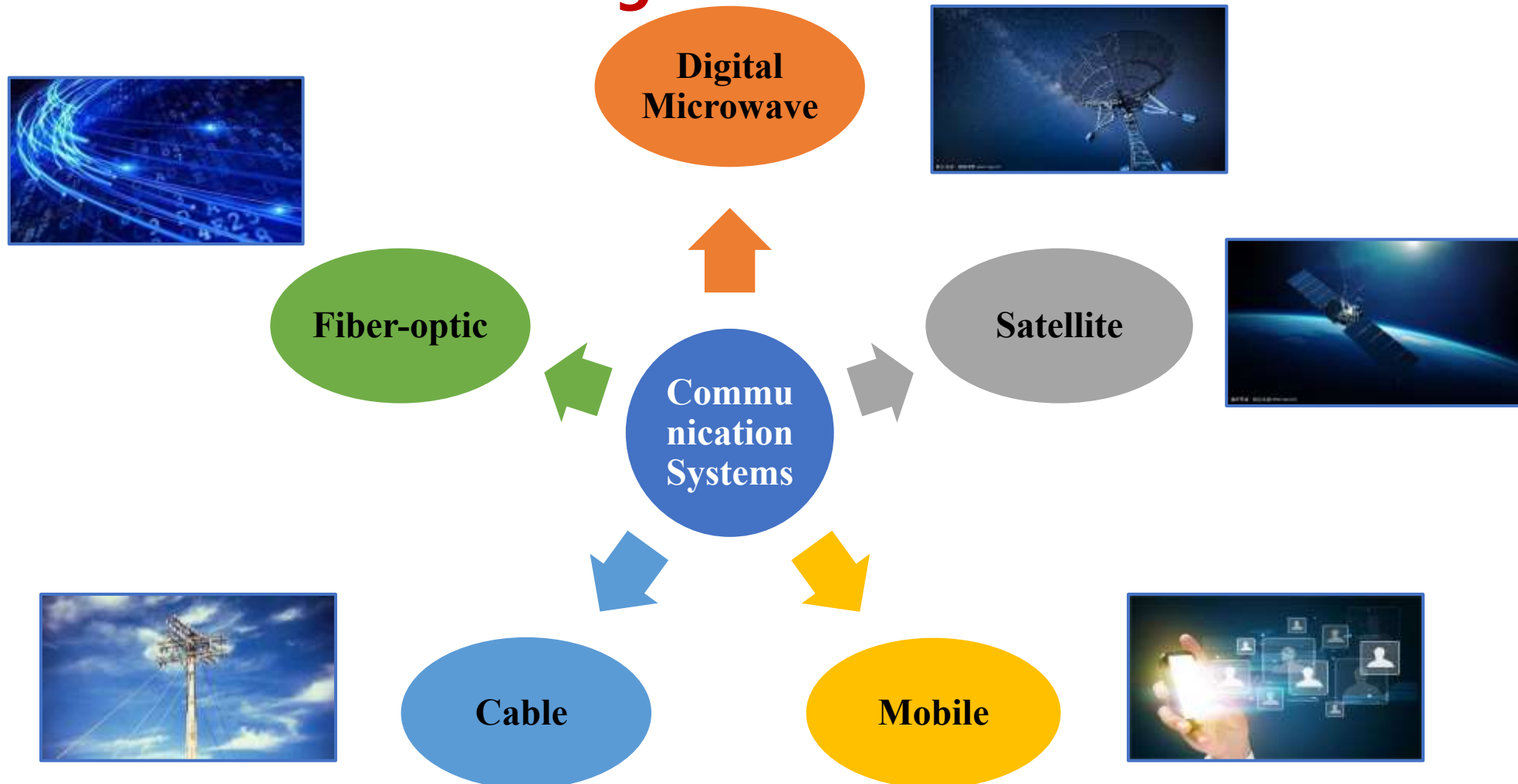
(1) Communication system: Telephone, transmitting electrical signals through wires



In 1876, Alexander Graham Bell invented the telephone, marking a new era in communication

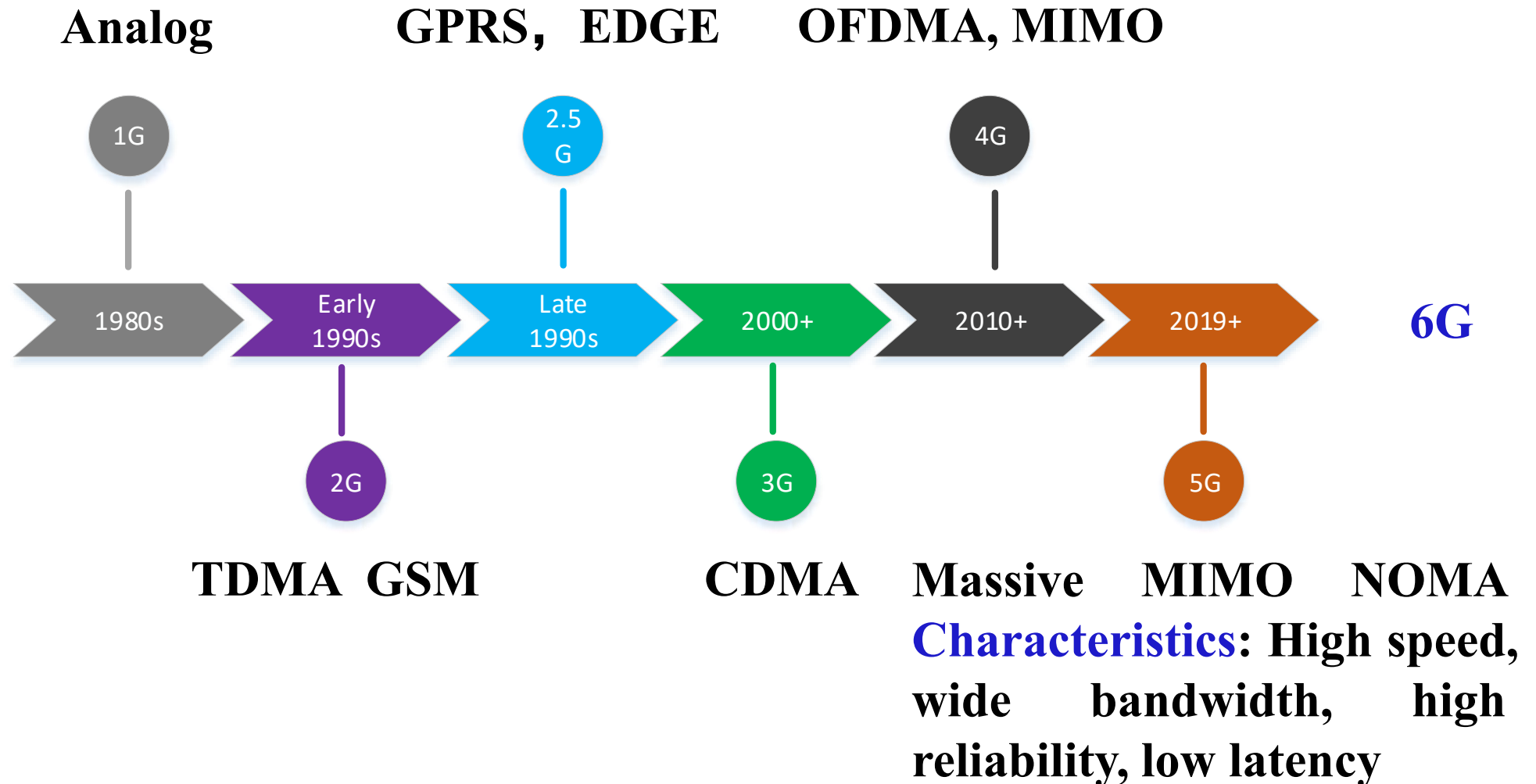
1. Basic concepts of signals and systems

(1) Communication system: Evolution of Modern Communication Technologies



1. Basic concepts of signals and systems

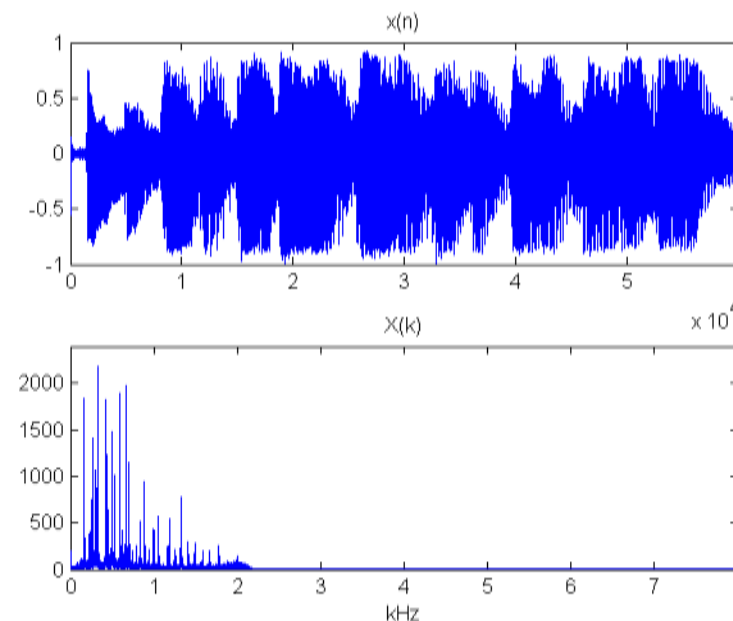
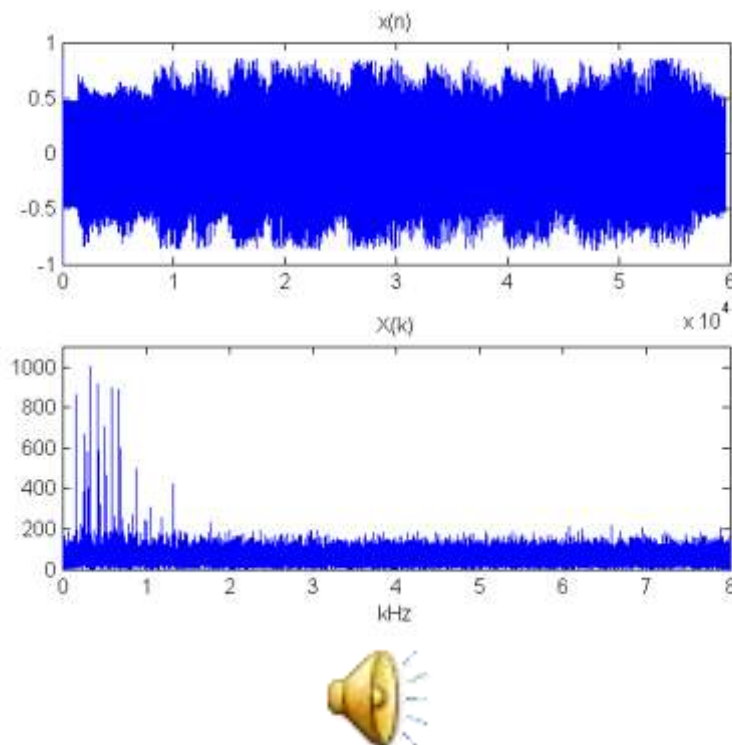
(1) Communication system: Mobile Communication



1. Basic concepts of signals and systems

(2) Signal Processing: Noise Reduction

- A segment of music mixed with noise, severely affecting listening quality
- Through **filtering processing**, the purpose of noise reduction is achieved



1. Basic concepts of signals and systems

(2) Signal Processing: Noise Reduction



Objective: To transform signals into forms that facilitate analysis and identification, enabling efficient estimation and selection of their characteristic parameters

1. Basic concepts of signals and systems

3. Learning method

Focus on the correspondence between **physical concepts** and **mathematical analysis**, do not calculate blindly

Pay attention to the **physical interpretation** of analysis results, the physical meaning when various parameters change, and the consequences they produce

The same problem may have **multiple solutions**; one should seek the simplest and most reasonable solution, and compare the advantages and disadvantages of each method

Even a considerable time after completing this course, it is still necessary to **repeatedly study** the basic concepts of this course

1. Basic concepts of signals and systems

4. Reference Book



Main Content

1. Signal Representation Methods

2. Description and Classification of Signals (by **Temporal Characteristics**)

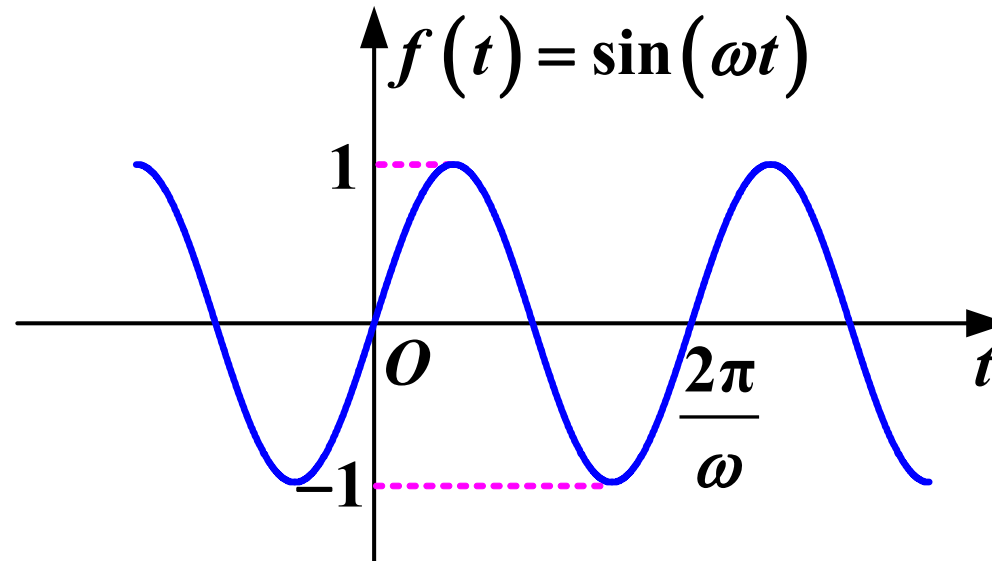
- (1) Deterministic Signals vs. Random Signals
- (2) Continuous Signals vs. Discrete Signals
- (3) Periodic Signals vs. Aperiodic Signals
- (4) Energy Signals vs. Power Signals
- (5) One-Dimensional Signals vs. Multidimensional Signals

2. Description and classification of signals

1. Methods for Signal Description

Signals can be represented as functions of one or more independent variables

- Graphical representation (waveform plotting)
- Tabular representation



2. Classification of Signals

By **Temporal Characteristics:**

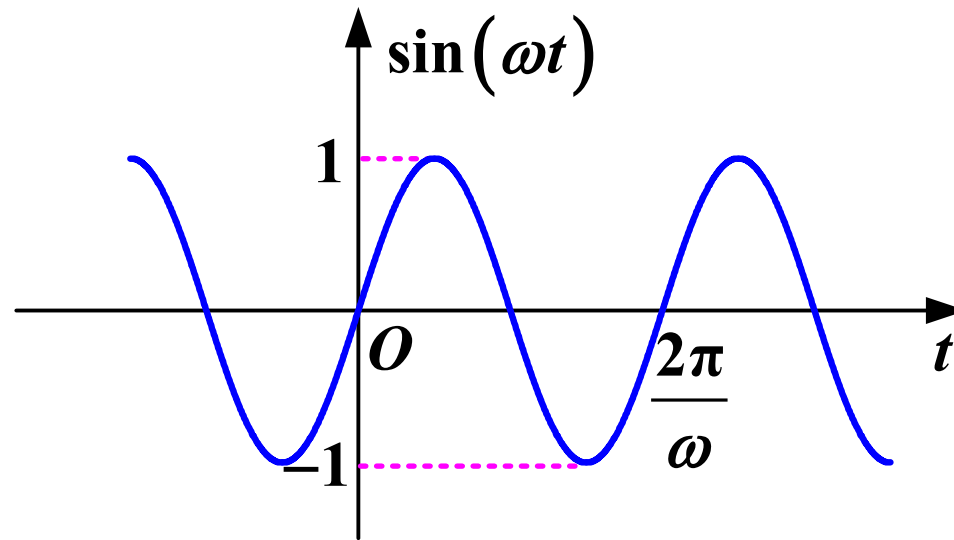
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2. Description and classification of signals

(1) Deterministic Signals vs. Random Signals

Deterministic Signals

For any given time instant t , the signal value $f(t)$ is precisely specified

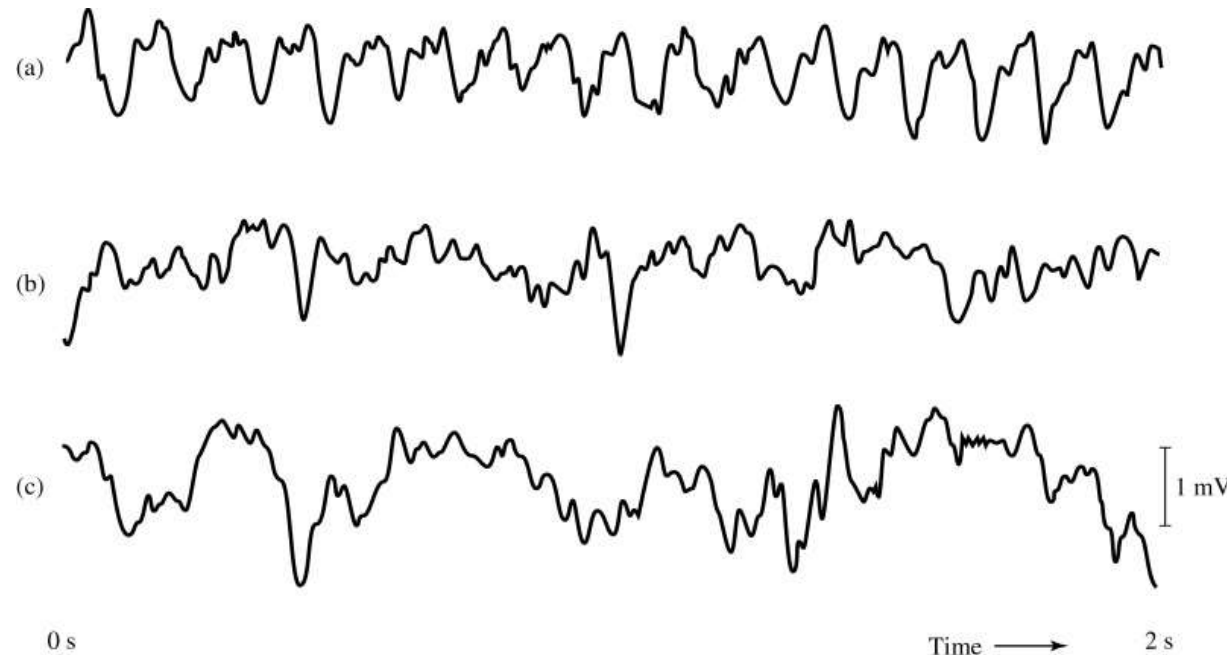


2. Description and classification of signals

(1) Deterministic Signals vs. Random Signals

Random Signals

Signals exhibiting unpredictable uncertainty, where instantaneous values cannot be precisely determined beforehand



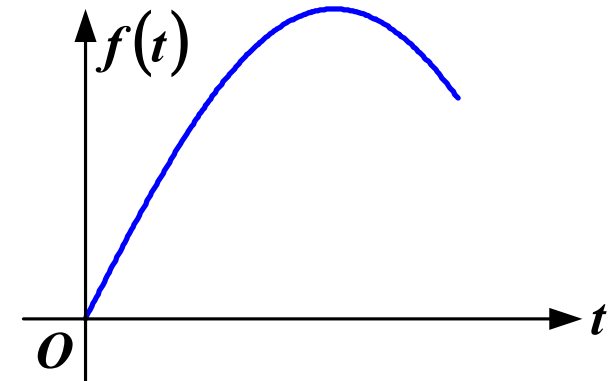
Rodent Electroencephalogram Signals

2. Description and classification of signals

(2) Continuous Signals vs. Discrete Signals

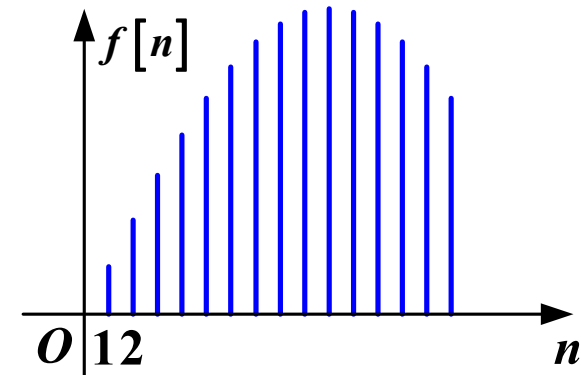
Continuous-Time Signals

Within the time domain of the signal's existence, the signal is defined at every instant (except for a finite number of discontinuity points), meaning a deterministic function value can be assigned at each point
Denoted by the continuous-time variable t



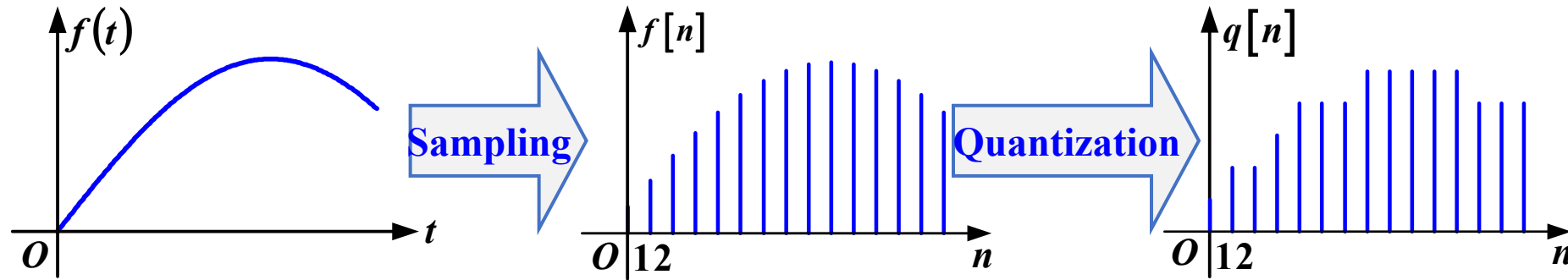
Discrete-Time Signals

Temporally discrete, these signals are defined only at specific, non-consecutive instants in time and are undefined at all other times
Denoted by the discrete-time variable n



2. Description and classification of signals

(2) Continuous Signals vs. Discrete Signals



Continuous Signal:

Time continuous

Amplitude continuous

Sampled Signal:

Time discrete

Amplitude continuous

Digital Signal:

Time discrete

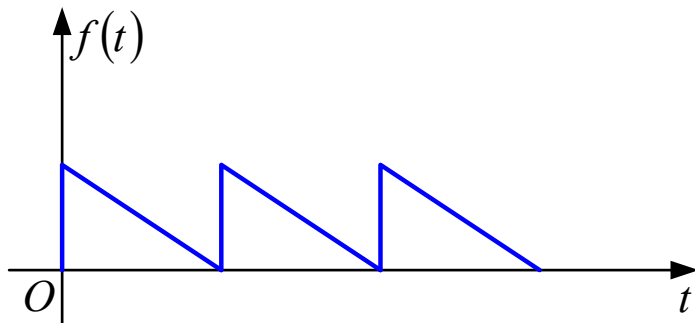
Amplitude discrete

2. Description and classification of signals

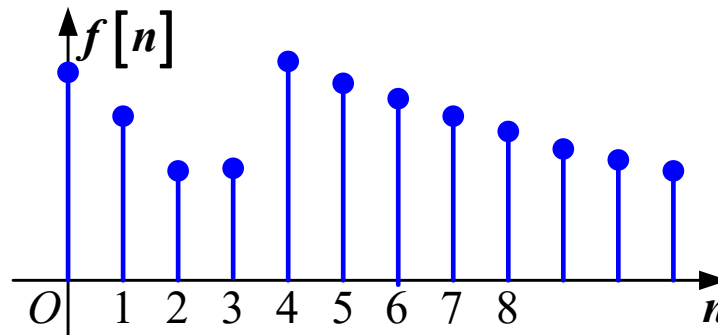
(2) Continuous Signals vs. Discrete Signals

Example 1:

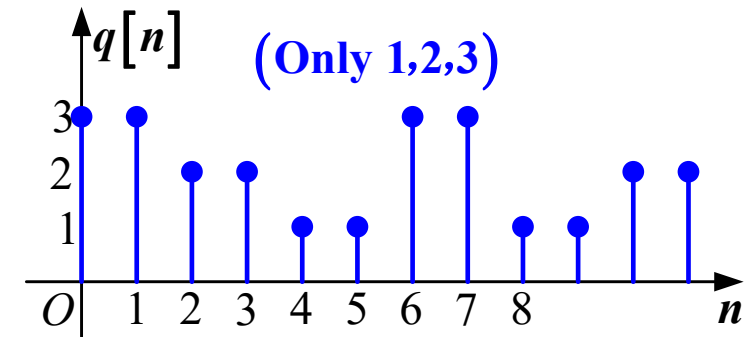
Determine whether the given waveform is a continuous-time signal or discrete-time signal, and if it is a discrete-time signal, identify whether it is a digital signal



Continuous Signal



Discrete Signal



Discrete Signal
Digital Signal

2. Description and classification of signals

(3) Periodic Signals vs. Aperiodic Signals

Periodic Signals

A signal that repeats its pattern indefinitely, with no beginning or end in its idealized form

$$f(t) = f(t + nT) \quad n = 0, \pm 1, \pm 2, \dots (\text{Any integer})$$

Fundamental Period (T): The **smallest positive value** T

Units: Seconds (s)

Fundamental Frequency (f_0)

$$f_0 = \frac{1}{T}$$

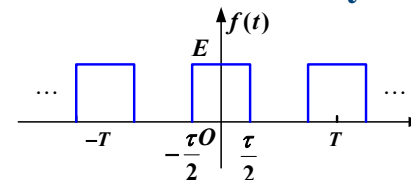
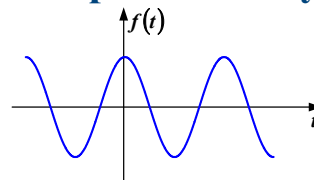
Units: Hertz (Hz)

Fundamental Angular Frequency (ω_0)

$$\omega_0 = 2\pi f_0 = \frac{2\pi}{T}$$

Units: Radians per second (rad/s)

Periodic Signals {
 Sinusoidal Periodic Signal (Harmonic Signal)
 Single frequency component (pure tone)
 Complex Periodic Signals (Any periodic signal not purely sinusoidal (e.g., square, triangle waves))
 Represented by Fourier series: Sum of harmonically related sinusoids



2. Description and classification of signals

(3) Periodic Signals vs. Aperiodic Signals

Example 2: Is the signal $f(t) = \cos 10t + \cos 30t$ periodic?

If yes, determine its fundamental period T and fundamental angular frequency ω_0

For $\cos 10t$: $T_1 = \frac{2\pi}{10}$

For $\cos 30t$: $T_2 = \frac{2\pi}{30}$

Check Periodicity Condition: A sum of periodic signals is periodic if and only if the ratio of their periods is a rational number

$$T = k_1 T_1 = k_2 T_2, \text{ where } k_1 \text{ and } k_2 \text{ are positive integers}$$

$$\frac{T_1}{T_2} = \frac{k_2}{k_1} = 3 \text{ is a rational number, therefore it is a periodic signal}$$

2. Description and classification of signals

(3) Periodic Signals vs. Aperiodic Signals

Example 2: Is the signal $f(t) = \cos 10t + \cos 30t$ periodic?

If yes, determine its fundamental period T and fundamental angular frequency ω_0

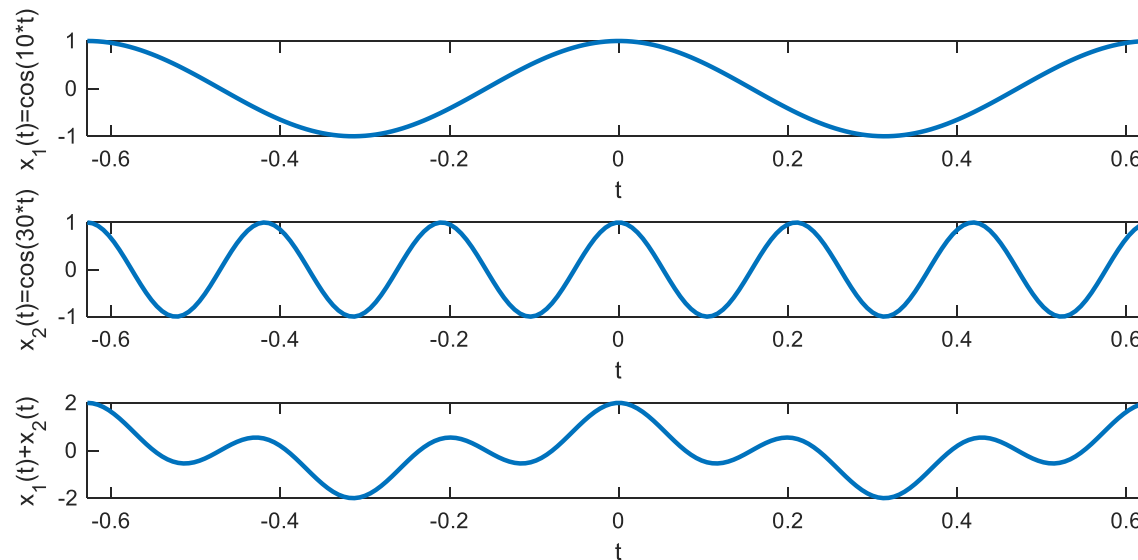
The minimum value of k_1 is 1

Fundamental Frequency

$$T = k_1 T_1 = \frac{2\pi}{10}$$

Fundamental Angular Frequency

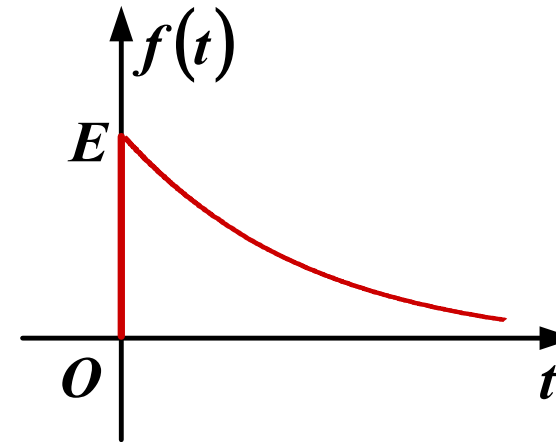
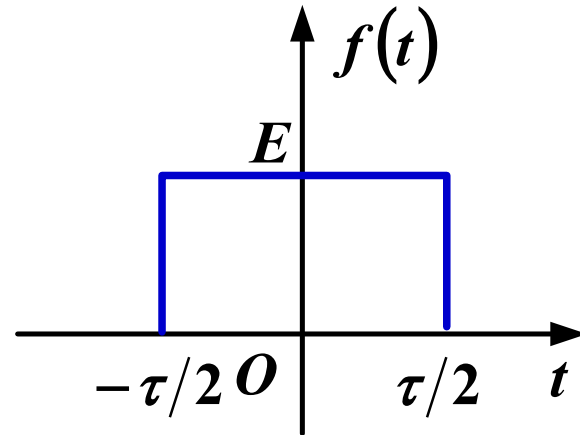
$$\omega = \frac{2\pi}{T} = 10$$



2. Description and classification of signals

(3) Periodic Signals vs. Aperiodic Signals

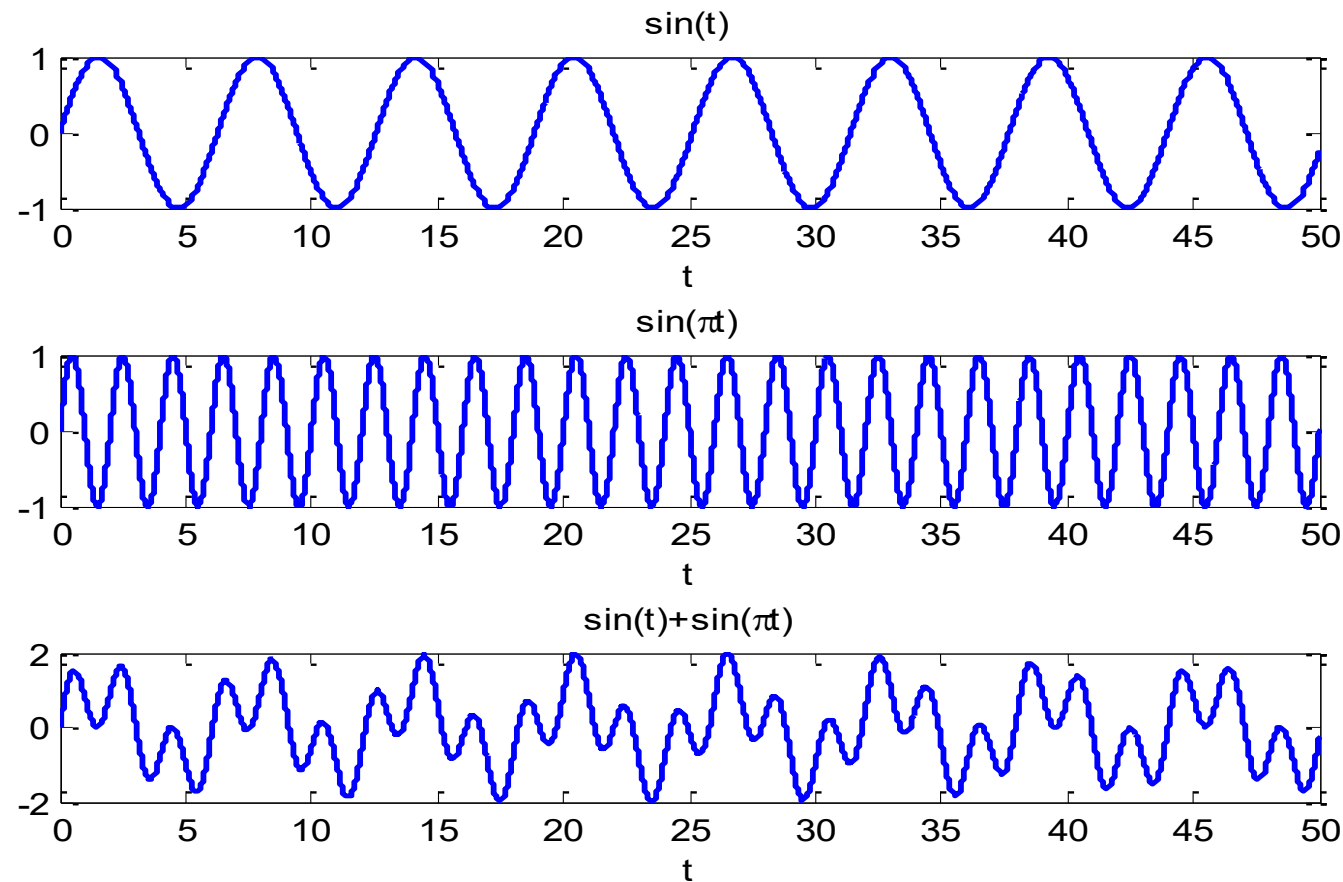
Aperiodic signals { Transient signals (Pulses, Decaying functions)
Quasi-periodic signals



2. Description and classification of signals

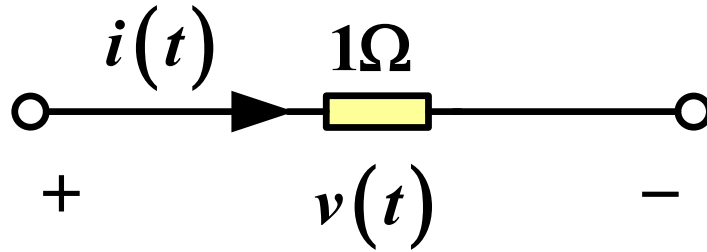
(3) Periodic Signals vs. Aperiodic Signals

Quasi-periodic signal: $\sin(\pi t) + \sin(t)$



2. Description and classification of signals

(4) Energy Signals vs. Power Signals



Normalized **instantaneous power**

$$p(t) = |f(t)|^2$$

Normalized energy of signal $f(t)$ (or simply signal **energy**)

The energy dissipated when signal voltage (or current) is applied to a 1Ω resistor

$$E = \int_{-\infty}^{\infty} |f(t)|^2 dt$$

Average Power of Signal:

The power dissipated by signal voltage (or current) across a 1Ω resistor. The average power over the entire time domain is:

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 dt$$

2. Description and classification of signals

(4) Energy Signals vs. Power Signals

Energy signal: A signal with finite total energy over the time interval $(-\infty, \infty)$ but zero average power

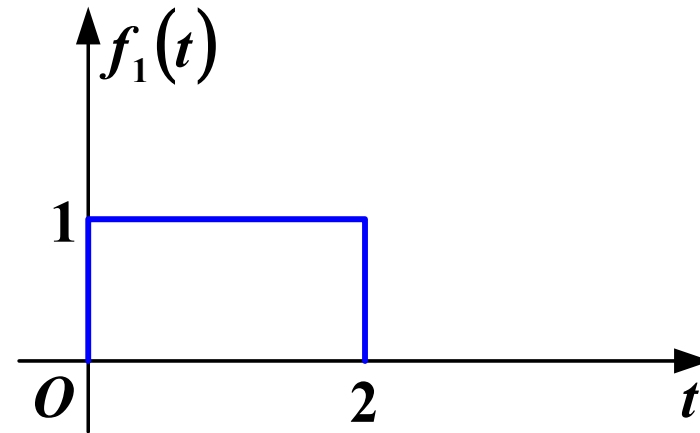
Most time-limited signals are energy signals

Example 3:

$$E_1 = \int_0^2 1^2 dt = 2$$

$$0 < E_1 < \infty$$

$$P_1 = 0$$



$f_1(t)$ is the energy signal

2. Description and classification of signals

(4) Energy Signals vs. Power Signals

Power signal: A signal with infinite total energy over the time interval $(-\infty, \infty)$ but finite average power

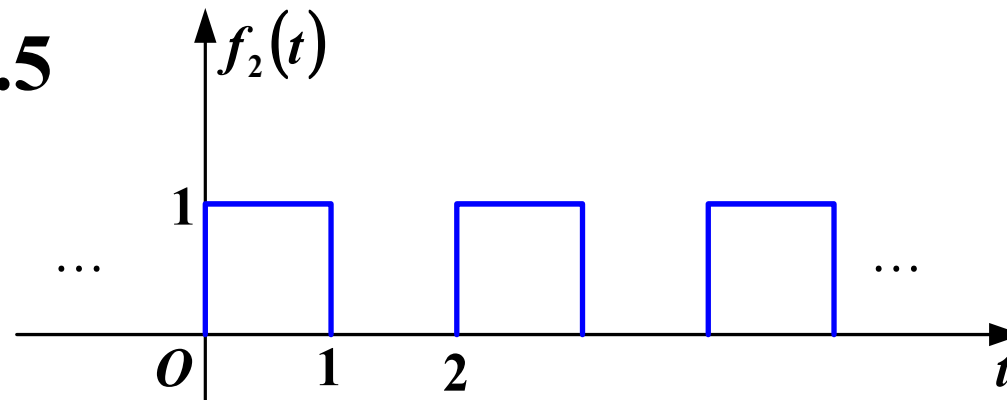
Examples include periodic signals, step functions, and signum functions

Example 4:

$$P_2 = \frac{1}{2} \int_0^1 1^2 dt = 0.5$$

$$0 < P_2 < \infty$$

$$E_2 = \infty$$

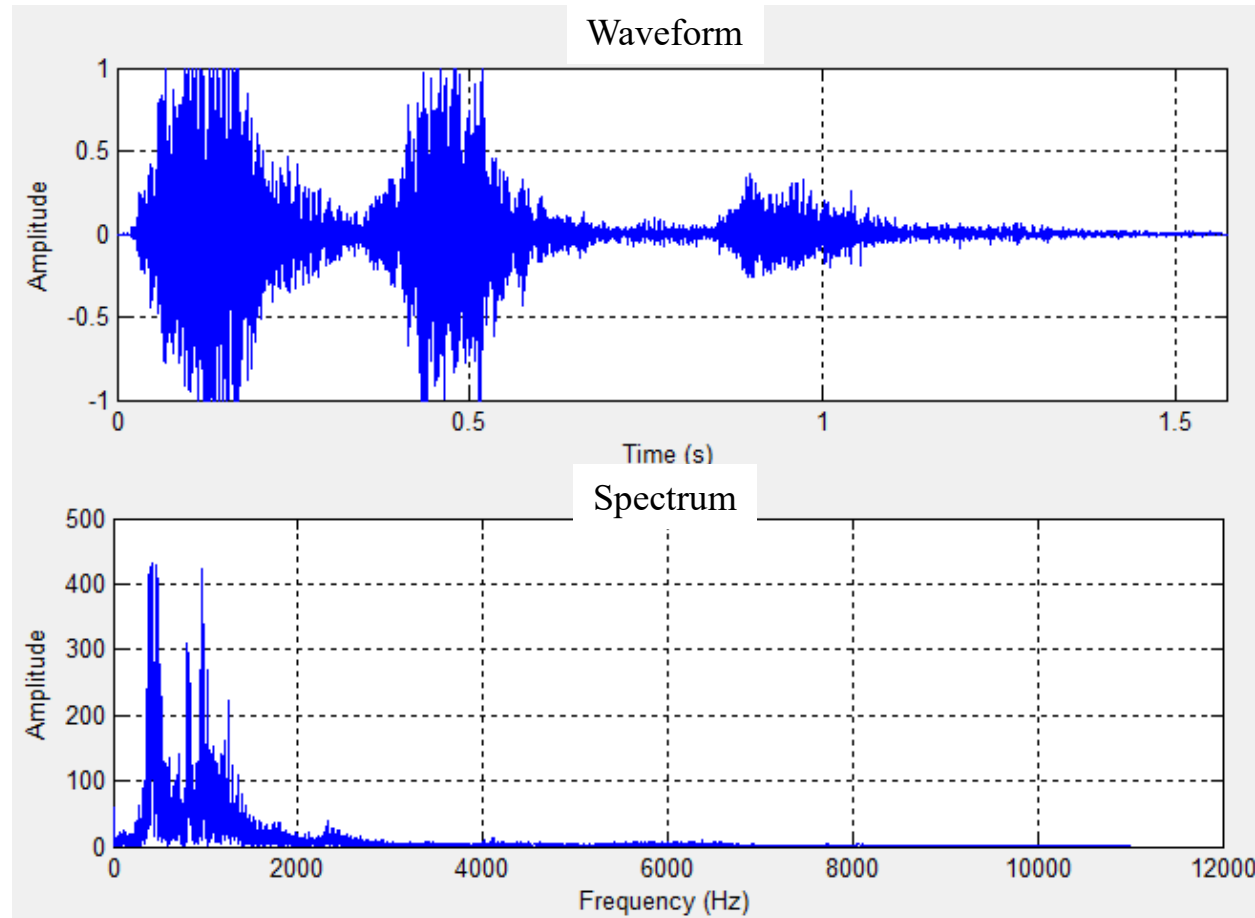


$f_2(t)$ is the power signal

2. Description and classification of signals

(5) One-Dimensional Signals vs. Multidimensional Signals

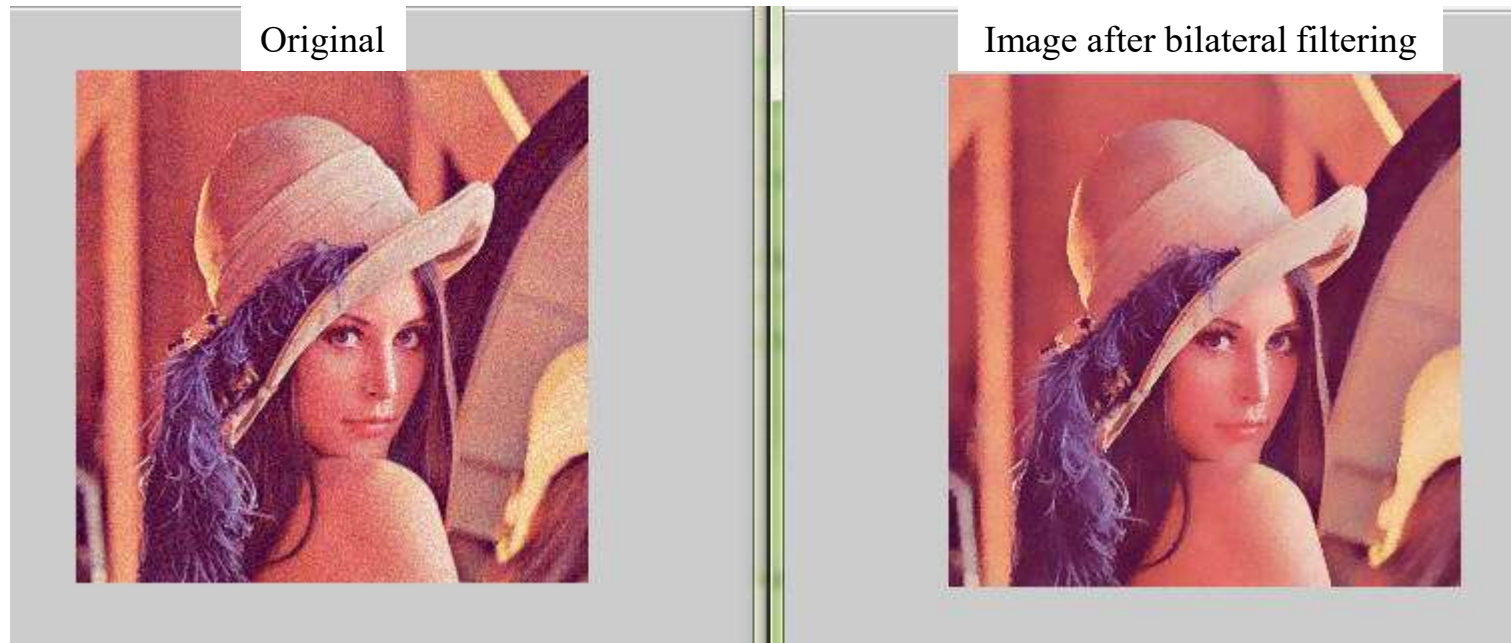
One-dimensional signal: A signal represented by a single independent variable (typically time or space), such as an audio signal



2. Description and classification of signals

(5) One-Dimensional Signals vs. Multidimensional Signals

Multidimensional signal: A signal requiring multiple independent variables for full characterization, such as image signals, electromagnetic field distributions, etc.



Lenna image, a widely used test image in image processing research

2. Description and classification of signals

(5) One-Dimensional Signals vs. Multidimensional Signals

Multidimensional Signals: Such as Wi-Fi signals. Luis Hernan, a Ph.D. student at Newcastle University in the UK, designed a device capable of capturing and visualizing Wi-Fi signals. The system continuously scans wireless networks and translates signal strength into dynamic displays of colored LEDs, with the intensity of each light band representing the corresponding signal strength



Summary

Signal Representation Methods

Description and Classification of Signals (by **Temporal Characteristics**)

- (1) Deterministic Signals vs. Random Signals
- (2) Continuous Signals vs. Discrete Signals
- (3) Periodic Signals vs. Aperiodic Signals
- (4) Energy Signals vs. Power Signals
- (5) One-Dimensional Signals vs. Multidimensional Signals

Main Content

1. Exponential signal
2. Sinusoidal signal
3. Complex exponential signal
4. Sampling signal
5. Gaussian pulse signal

3.1 Typical continuous signals

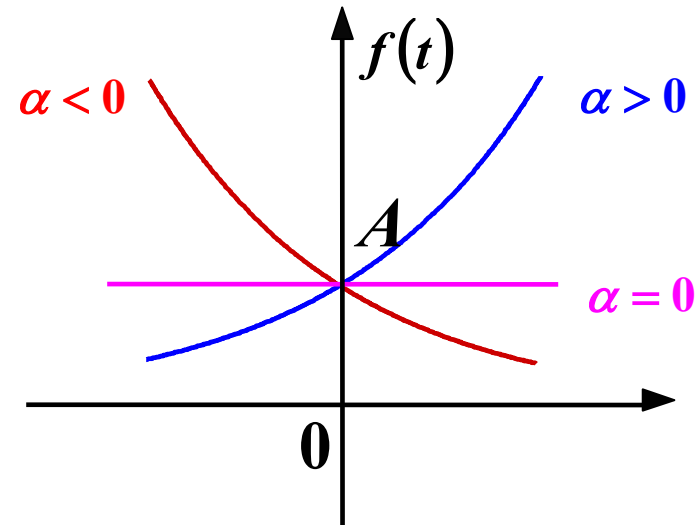
1. Exponential signal

$$f(t) = Ae^{\alpha t}$$

$\alpha = 0$ Constants, DC

$\alpha < 0$ ↘

$\alpha > 0$ ↗



Key property: Their differentiation and integration with respect to time still yield **exponential** forms

3.1 Typical continuous signals

1. Exponential signal

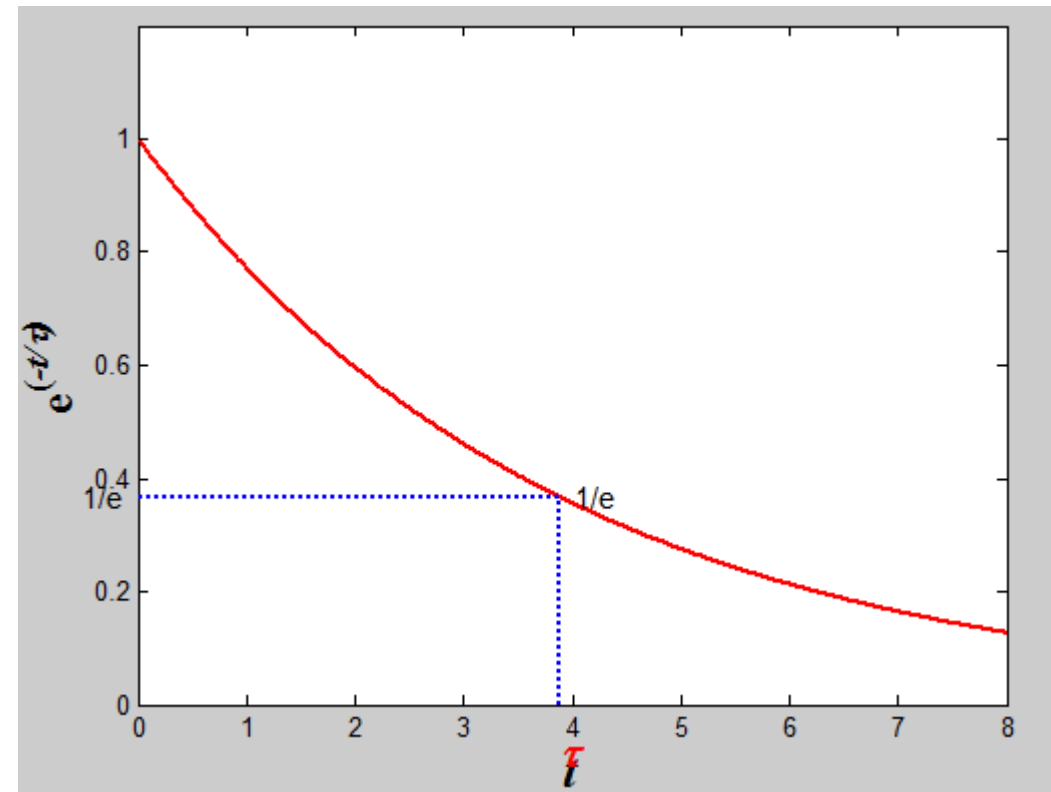
- **Single-sided exponential signal**

$$f(t) = e^{-\frac{t}{\tau}} \quad (t > 0)$$

- τ , time constant
- Has the dimension of time
- Represents the signal decay rate

When $t = \tau$

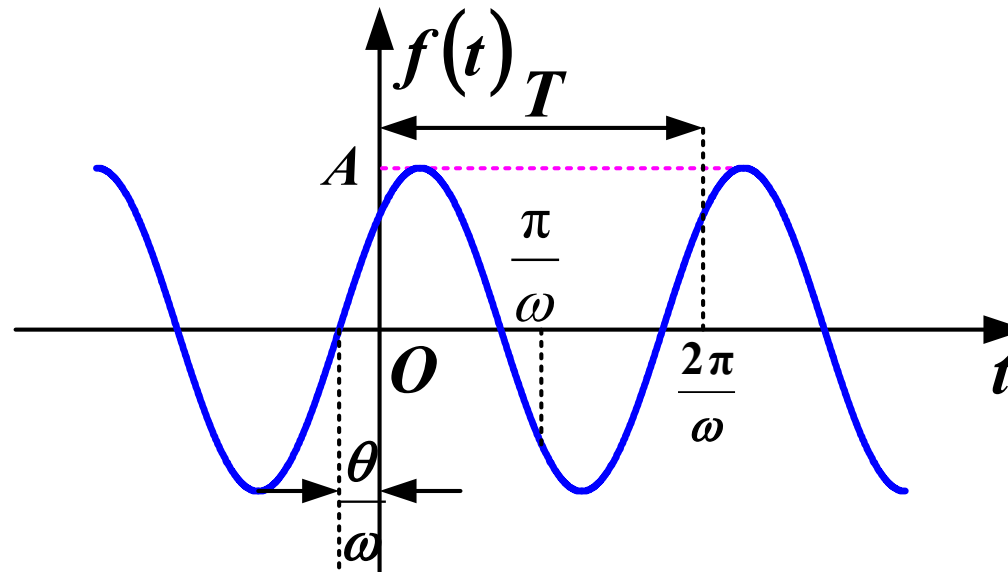
$$f(t) = \frac{1}{e}$$



3.1 Typical continuous signals

2. Sinusoidal signal

$$f(t) = A \sin(\omega t + \theta)$$



Amplitude: A

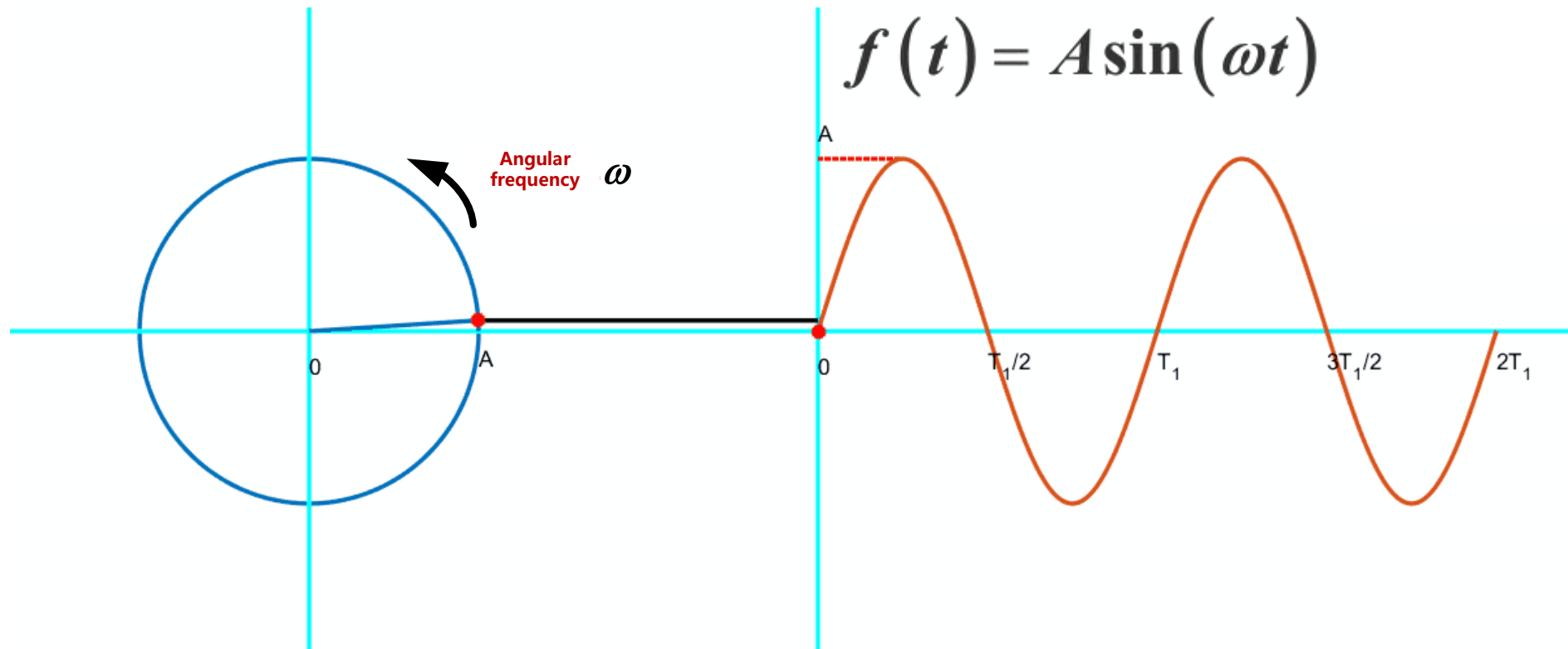
Angular frequency: ω

Initial phase: θ

3.1 Typical continuous signals

2. Sinusoidal signal

Angular Frequency: ω **Frequency:** $f = \frac{\omega}{2\pi}$ **Period:** $T = \frac{1}{f}$



3.1 Typical continuous signals

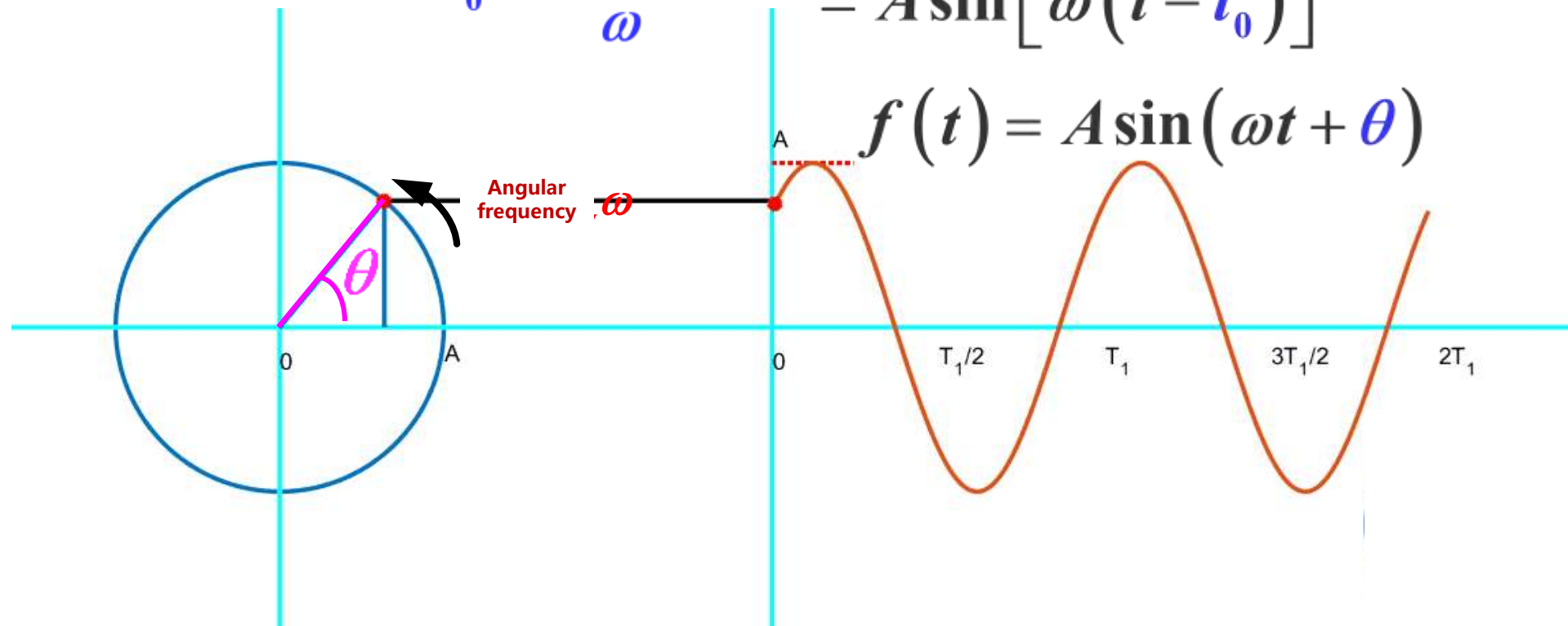
2. Sinusoidal signal

- Initial phase

$$f(t) = A \sin(\omega t + \theta) = A \sin \left[\omega \left(t + \frac{\theta}{\omega} \right) \right]$$

Let $t_0 = -\frac{\theta}{\omega}$

$$= A \sin \left[\omega (t - t_0) \right]$$



2. Sinusoidal signal

- Euler's Formula



Leonhard Euler (1707–1783), Swiss mathematician and natural scientist.

Widely regarded as one of the four greatest mathematicians in history, alongside Archimedes, Newton, and Gauss.

He introduced numerous mathematical notations, including:

- π (1736)
- i (imaginary unit, 1777)
- e (base of natural logarithm, 1748)
- $\sin$ and $\cos$ (1748)
- $\tan$ (1753)
- Δx (increment notation, 1755)
- Σ (summation symbol, 1755)
- $f(x)$ (function notation, 1734)

(Note: Euler's formula, $e^{i\theta} = \cos\theta + i\sin\theta$, is hailed as the "bridge of mathematics" for its profound connection between exponential and trigonometric functions, with far-reaching impacts in signal analysis, quantum mechanics, and beyond.)

3.1 Typical continuous signals

2. Sinusoidal signal

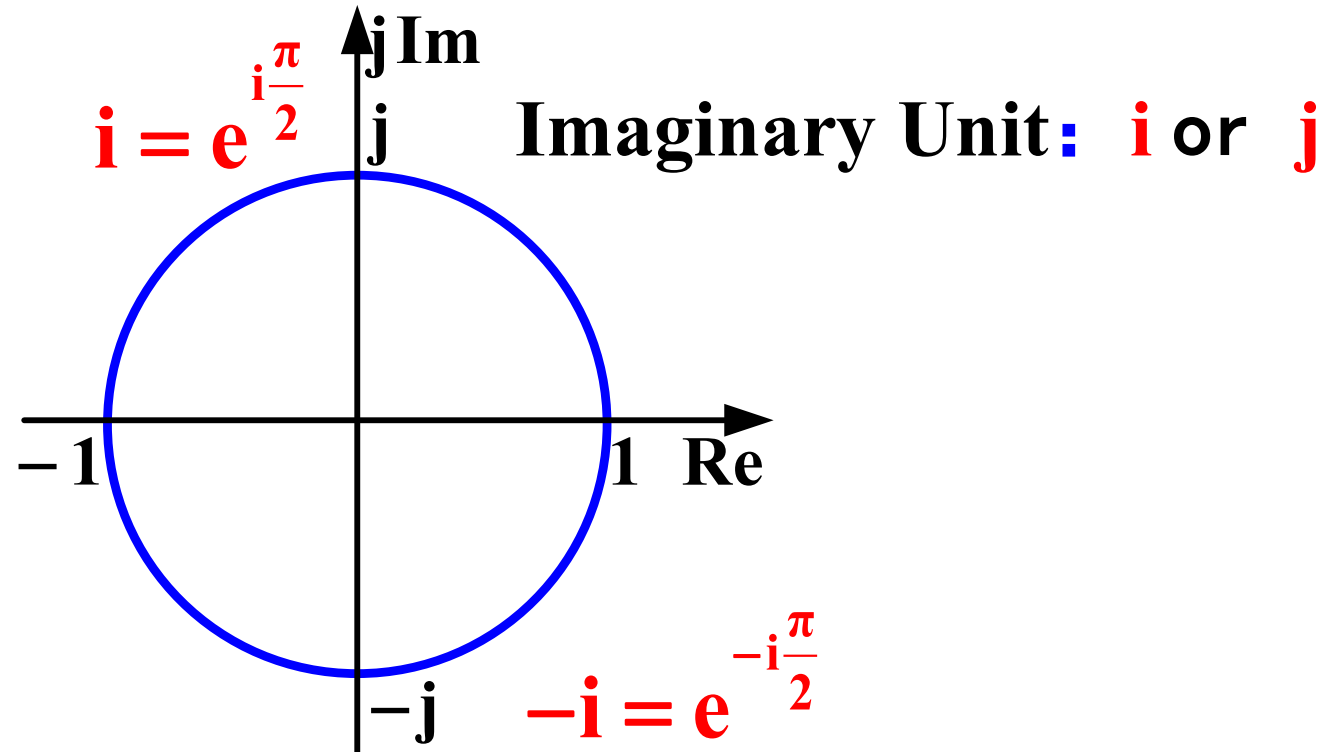
- Euler's Formula

$$-1 = e^{i\pi} = e^{i(-\pi)}$$

Euler's identity

$$e^{i\pi} + 1 = 0$$

God's Formula



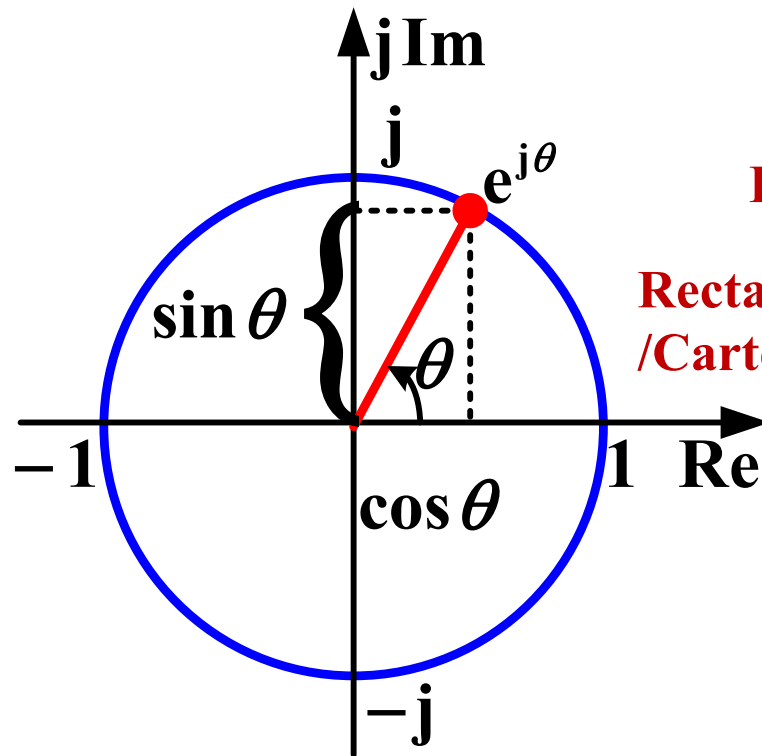
Complex Plane

3.1 Typical continuous signals

2. Sinusoidal signal

- **Euler's Formula**

On the complex plane, a ray from the origin to a point on the unit circle, forming an angle θ with the real axis, can be expressed as



Polar Form: $1 \angle \theta = e^{j\theta}$

Rectangular / Cartesian Form: $\cos \theta + j \sin \theta$

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$$e^{-j\theta} = \cos \theta - j \sin \theta$$

3.1 Typical continuous signals

2. Sinusoidal signal

- Euler's Formula

Let $\theta = \omega t$

$$\begin{cases} e^{j\omega t} = \cos(\omega t) + j\sin(\omega t) \\ e^{-j\omega t} = \cos(\omega t) - j\sin(\omega t) \end{cases}$$

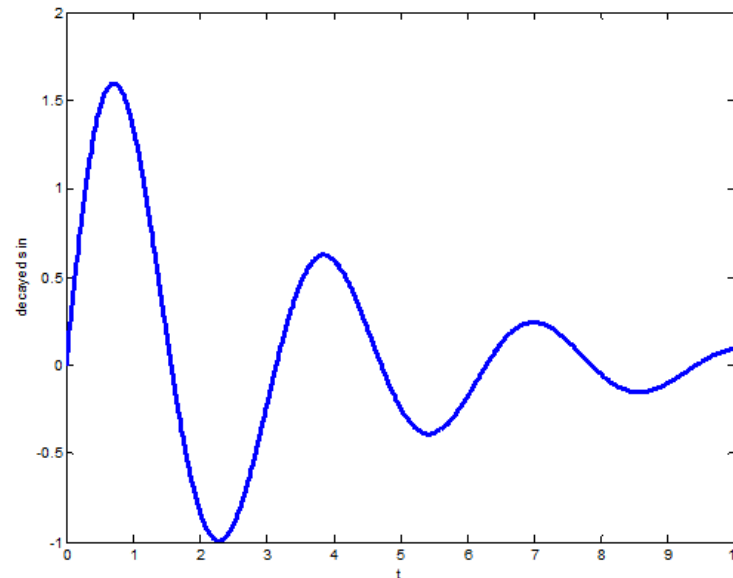
$$\begin{cases} \cos(\omega t) = \frac{1}{2}(e^{j\omega t} + e^{-j\omega t}) \\ \sin(\omega t) = \frac{1}{2j}(e^{j\omega t} - e^{-j\omega t}) \end{cases}$$

The Bridge Between Complex Exponential Functions and Trigonometric Functions

2. Sinusoidal signal

- **Single-Sided Damped Sinusoidal Signal**

$$f(t) = \begin{cases} K e^{-\alpha t} \sin(\omega t) & t \geq 0 \\ 0 & t < 0 \end{cases} \quad (\alpha > 0)$$



3.1 Typical continuous signals

3. Complex exponential signal

$$f(t) = Ke^{st} \quad (-\infty < t < \infty)$$

- $s = \sigma + j\omega$ is a complex number, known as complex frequency
- σ, ω are all constants
- σ , the dimension is $1/s$, ω , the dimension is rad/s

$$\omega = 0$$

$$f(t) = Ke^{\sigma t}$$

Exponential signal

3.1 Typical continuous signals

3. Complex exponential signal

$$\omega \neq 0$$

$$f(t) = Ke^{st}$$

$$= Ke^{(\sigma + j\omega)t}$$

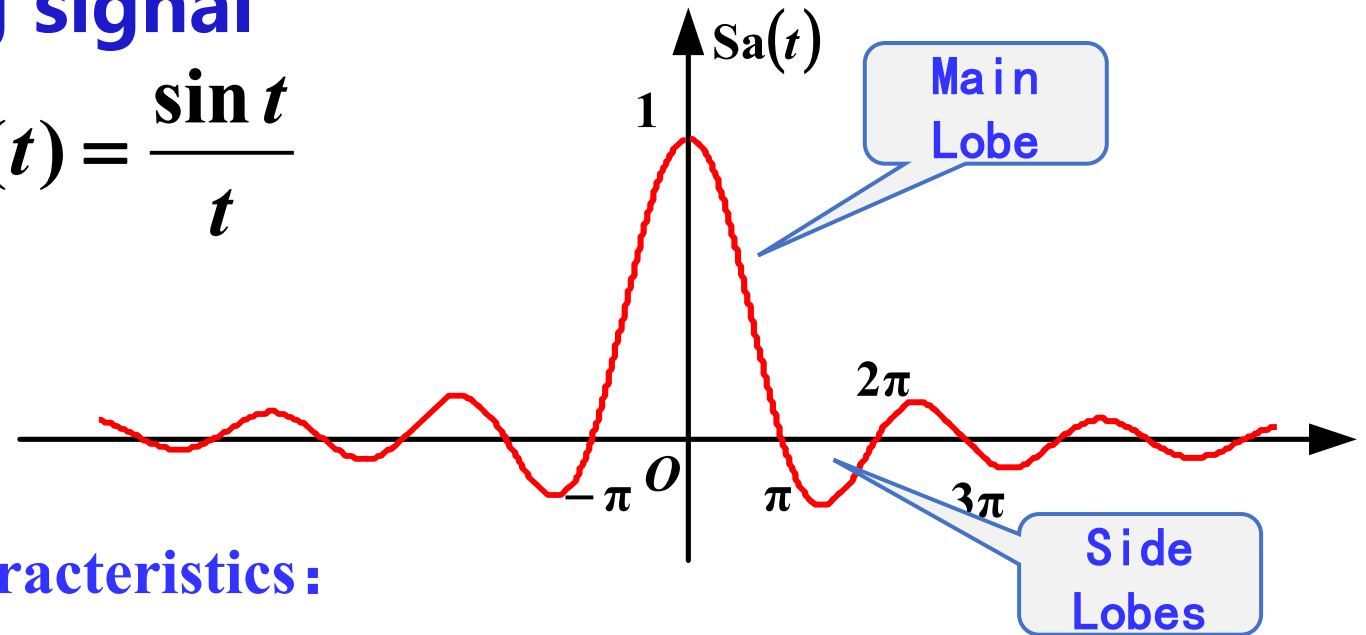
$$= Ke^{\sigma t} \cos(\omega t) + jKe^{\sigma t} \sin(\omega t)$$

$$\text{Real and Imaginary Parts} \left\{ \begin{array}{ll} \sigma = 0, & \text{Constant-Amplitude} \\ \sigma > 0, & \text{Amplifying} \\ \sigma < 0, & \text{Decaying} \end{array} \right\} \text{Oscillation}$$

3.1 Typical continuous signals

4. Sampling signal

$$Sa(t) = \frac{\sin t}{t}$$



Characteristics:

$$Sa(-t) = Sa(t), \text{ Even Function}$$

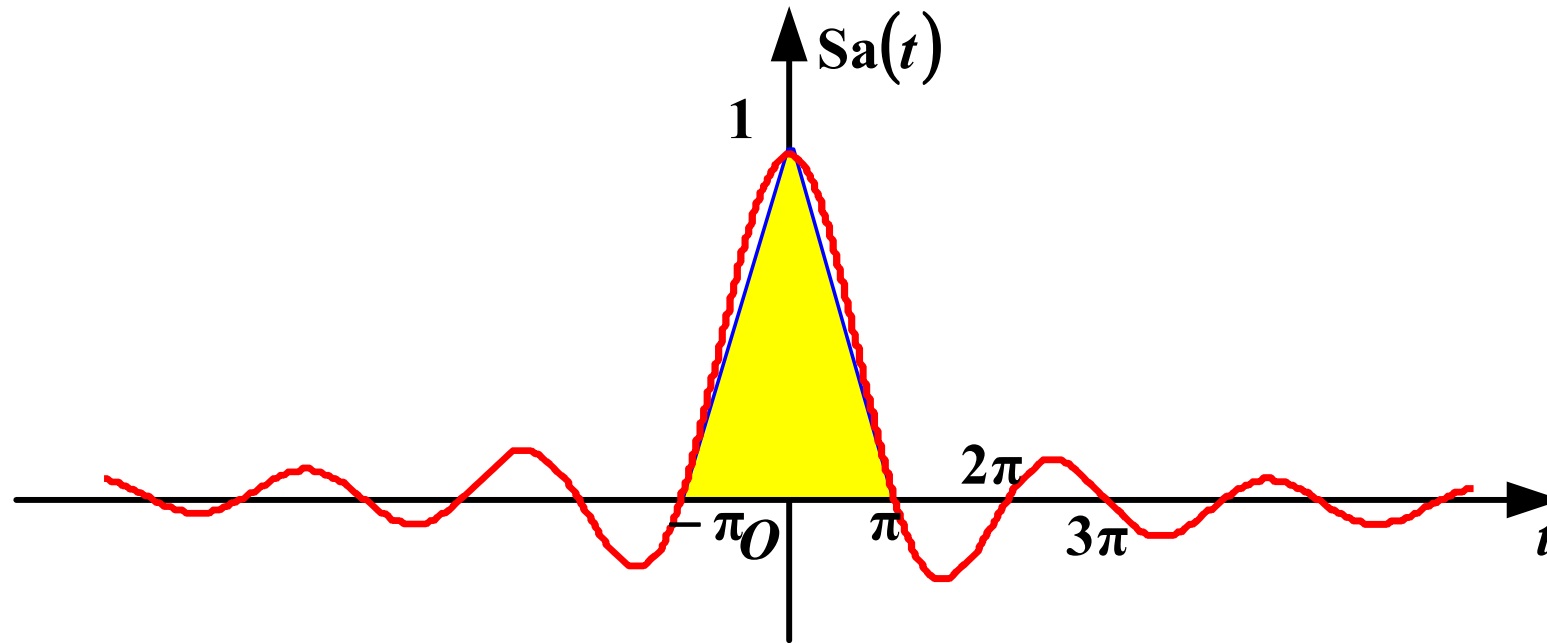
$$t = 0, Sa(t) = 1, \text{ thus } \lim_{t \rightarrow 0} Sa(t) = 1$$

$$Sa(t) = 0, \quad t = \pm n\pi, \quad n = 1, 2, 3 \dots$$

$$\lim_{t \rightarrow \pm\infty} Sa(t) = 0$$

3.1 Typical continuous signals

4. Sampling signal



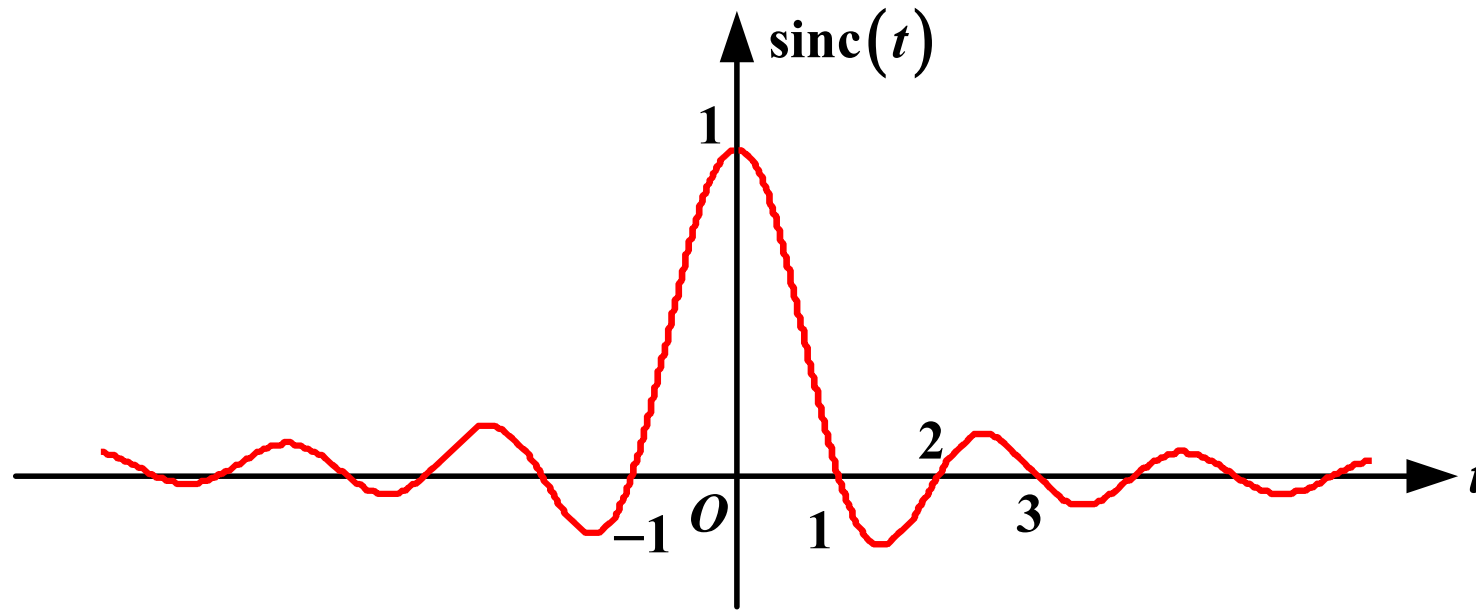
$$\int_0^{\infty} \text{Sa}(t) dt = \frac{\pi}{2}$$

$$\int_{-\infty}^{\infty} \text{Sa}(t) dt = \pi$$

Precisely equals
the area of the
inscribed triangle

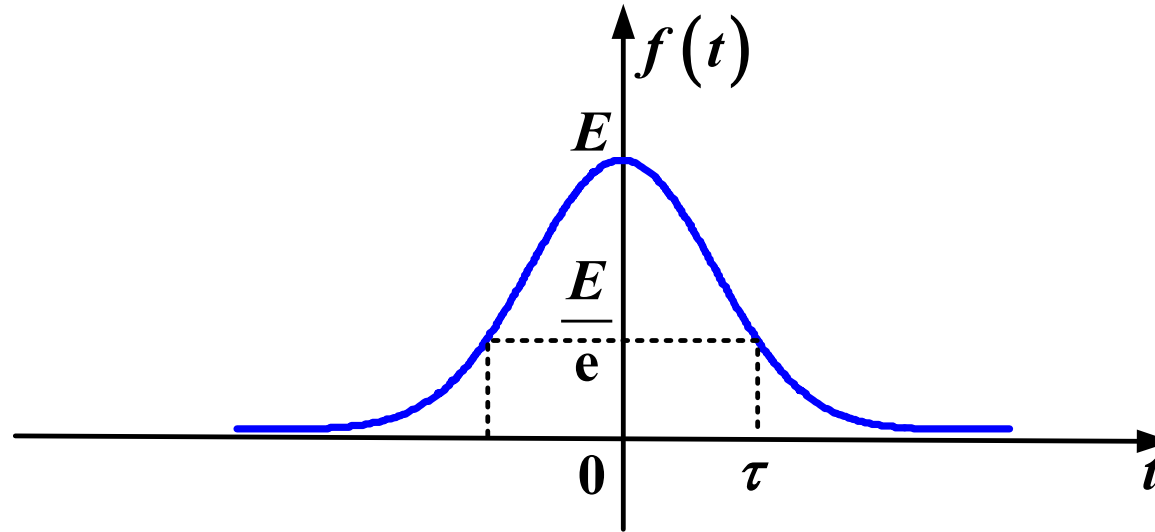
3.1 Typical continuous signals

4. Sampling signal



$$\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$$

5. Gaussian pulse signal



$$f(t) = E e^{-\left(\frac{t}{\tau}\right)^2}$$

Summary

1. Exponential signal
2. Sinusoidal signal
3. Complex exponential signal
4. Sampling signal
5. Gaussian pulse signal

Main Content

1. Dependent Variable Operations

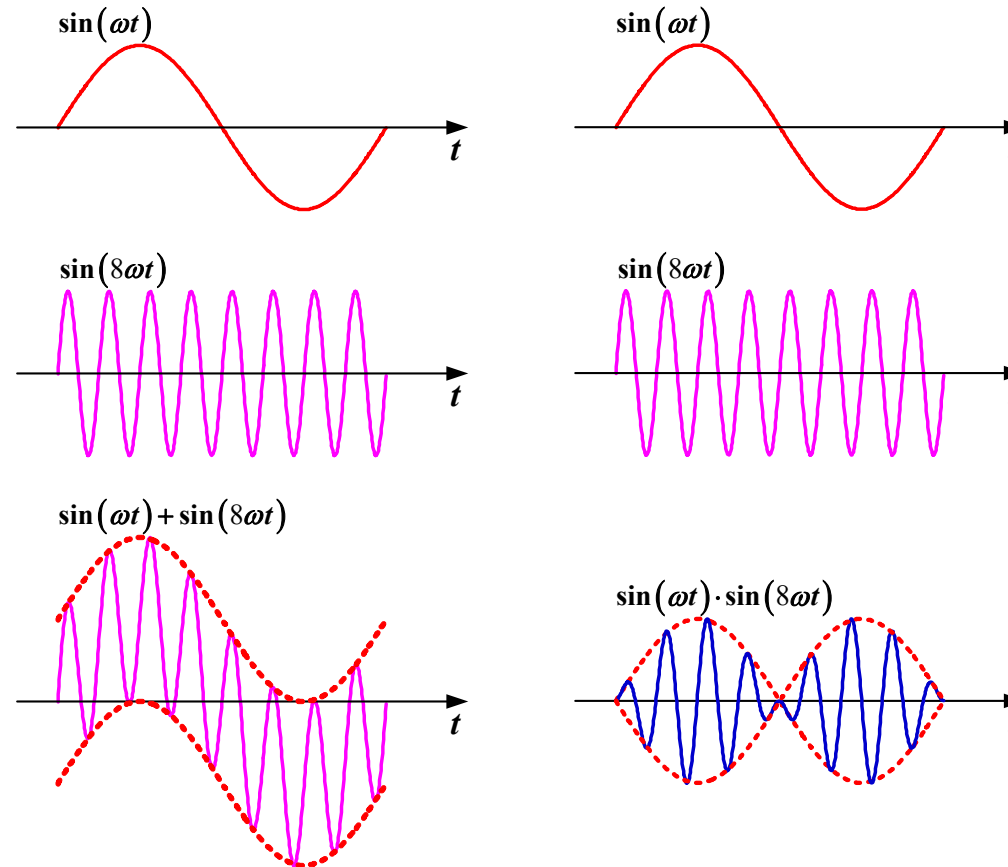
- Addition and Multiplication
- Differentiation and Integration

2. Independent Variable Transformations

- Time Shifting (Translation)
- Time Reversal
- Time Scaling

1. Addition and Multiplication

The operations where two signals are combined by adding or multiplying their corresponding values at each identical time instant



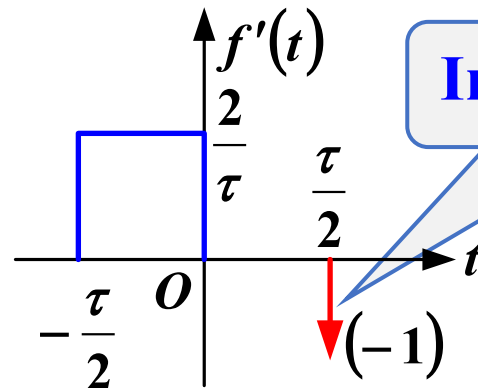
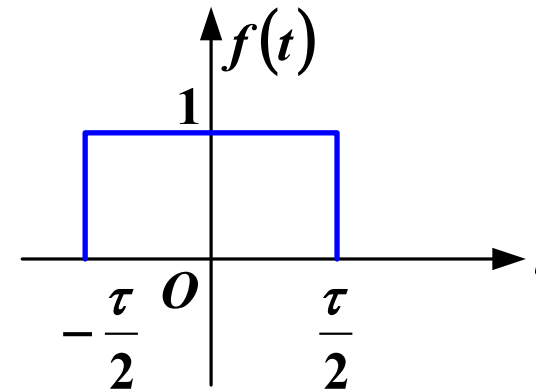
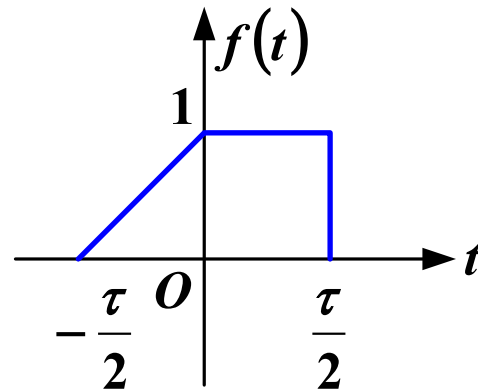
3.2 Operations on continuous-time signals

2. Differentiation and Integration

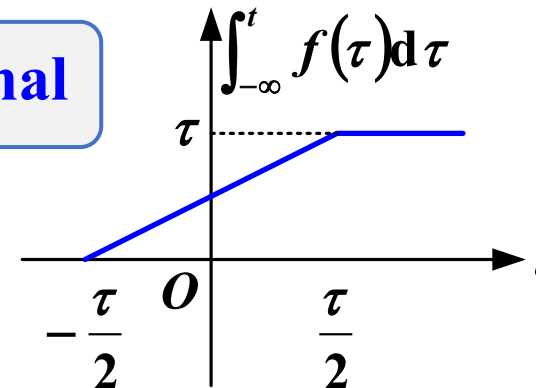
Differentiation: $f'(t) = \frac{d f(t)}{d t}$

Integration: $\int_{-\infty}^t f(\tau) d\tau$

Running
integral



Impulse Signal



3. Independent Variable Transformations

Including shifting, reversal, and scaling transformations

$$f(t) \rightarrow f(at + b)$$

Where

f : represents a function, a mapping

$at + b$: argument

t : independent variable

After transformation, the horizontal coordinate t of the waveform remains **unchanged**

(1) Shifting

$$f(t) \rightarrow f(t - \tau)$$

Translate the signal $f(t)$ along the t -axis to obtain the shifted signal $f(t-\tau)$, where τ is a constant

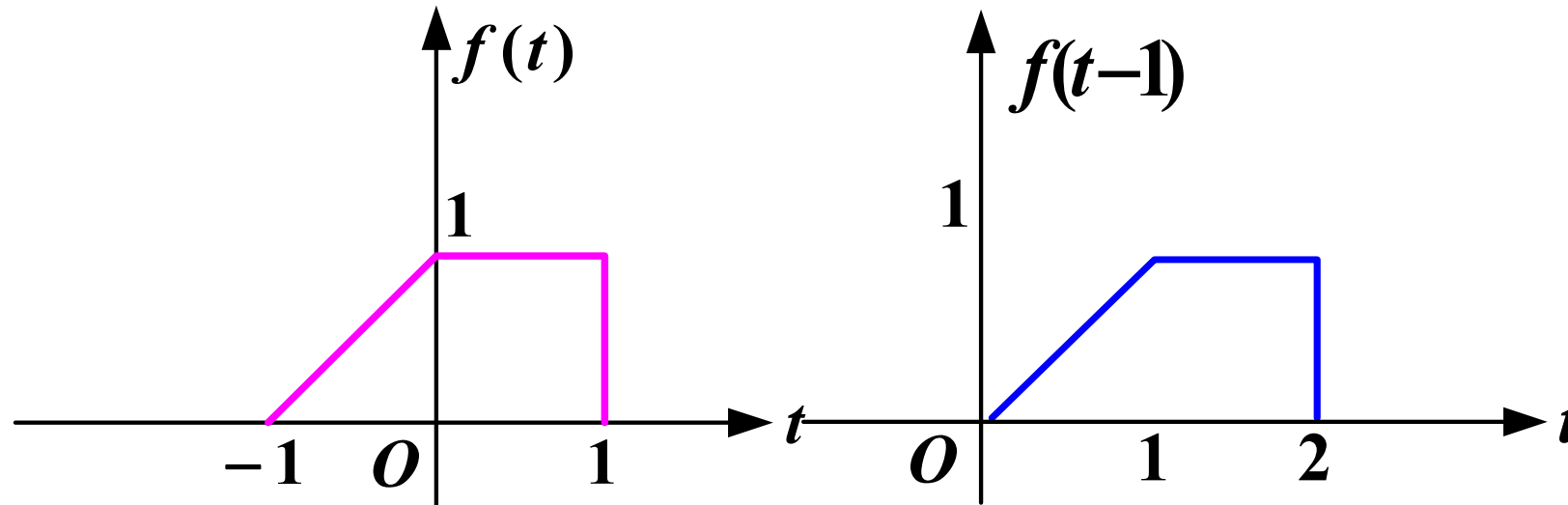
$\tau > 0$, Right shift (delay)

$\tau < 0$, Left shift (advance)

3.2 Operations on continuous-time signals

Example 1

Given the waveform of $f(t)$, plot the waveform for $f(t-1)$



Preserve amplitudes, Identical outputs, Shifting time coordinates

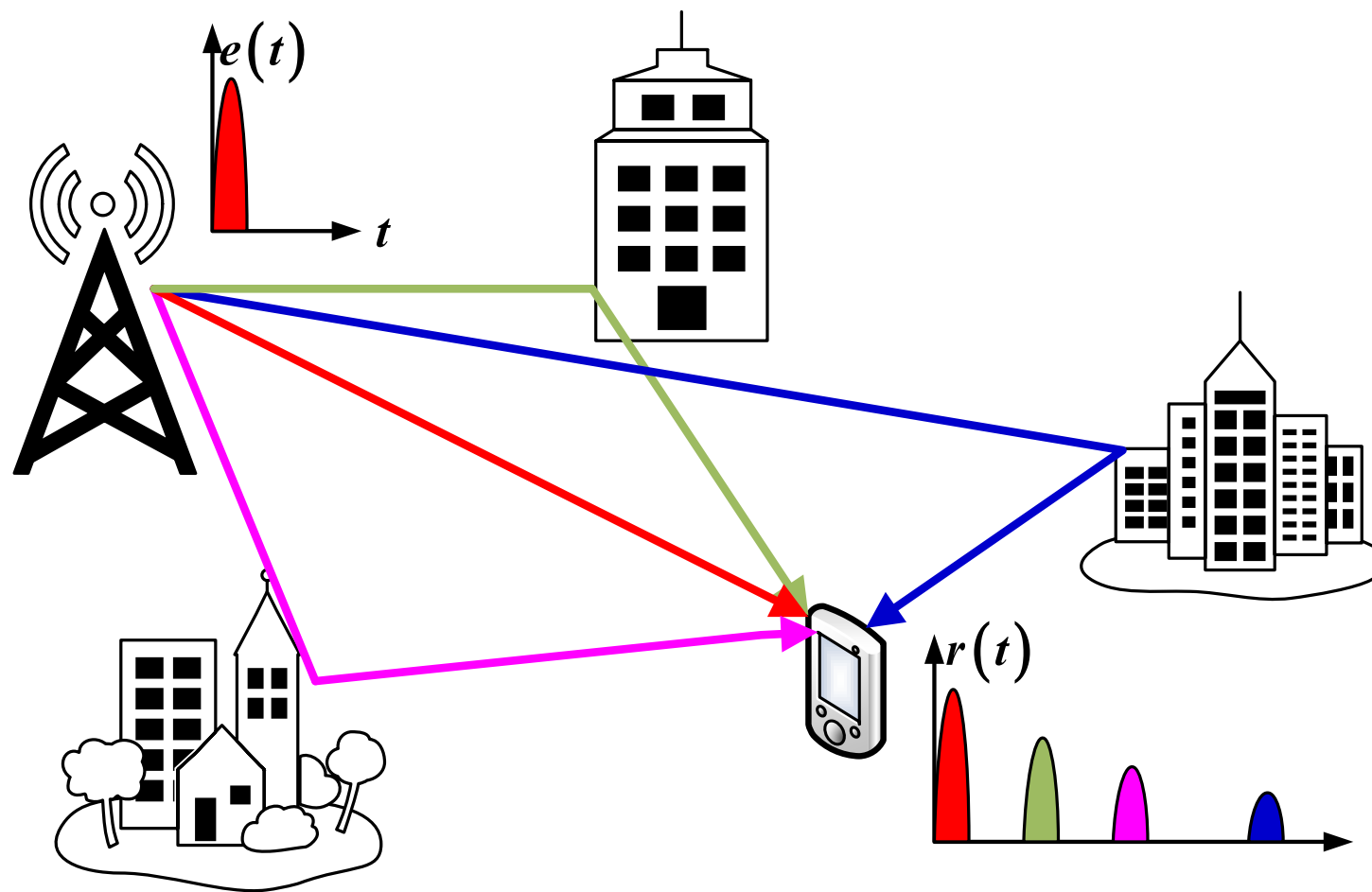
$$\begin{cases} t = 0 \\ f(t) = 1 \end{cases}$$

$$\begin{cases} t - 1 = 0 \\ f(t - 1) = 1 \end{cases}$$

$$\begin{cases} t = 1 \\ f(t - 1) = 1 \end{cases}$$

3.2 Operations on continuous-time signals

Signal Delay Example – Multipath Transmission



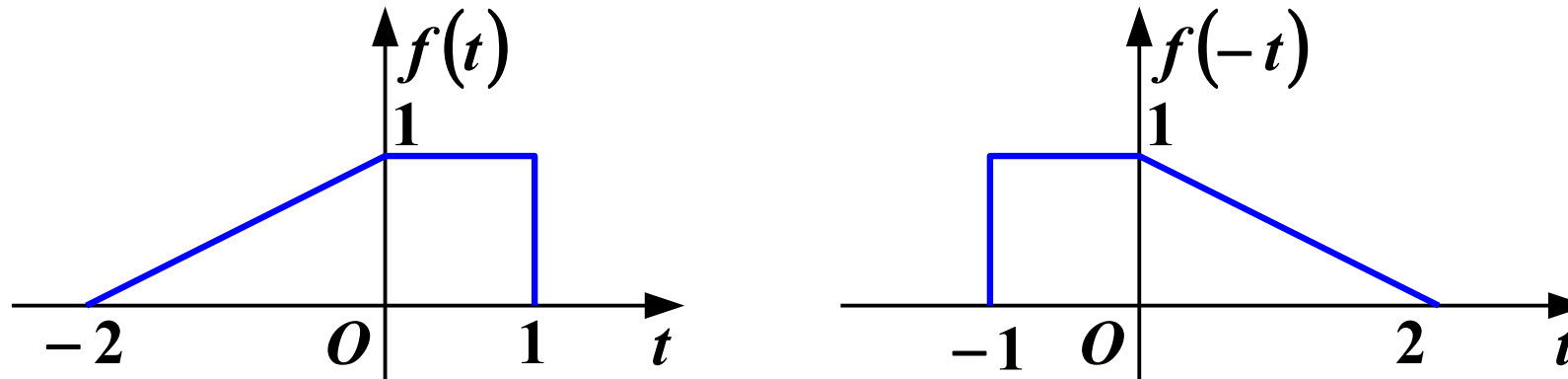
$$r(t) = a_1 e(t - \tau_1) + a_2 e(t - \tau_2) + a_3 e(t - \tau_3) + a_4 e(t - \tau_4)$$

3.2 Operations on continuous-time signals

(2) Reversal

Fold the signal along the vertical axis, swapping its past and future

$$f(t) \rightarrow f(-t)$$



In reality, no physical device can achieve this operation. However, this concept can be implemented in digital signal processing—for example, using a last-in-first-out (LIFO) stack

(3) Scaling

Waveform Compression and Expansion

$$f(t) \rightarrow f(at)$$

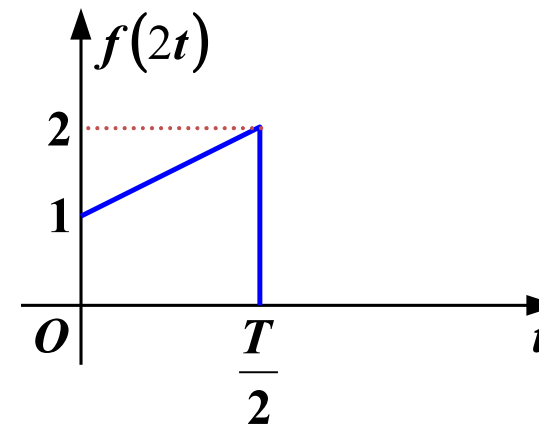
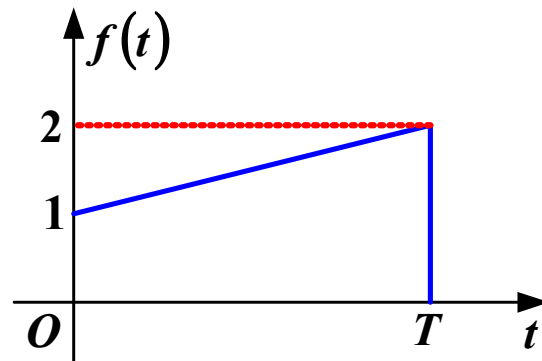
a is a real-valued scaling factor

3.2 Operations on continuous-time signals



Example 2

Given $f(t)$, plot $f(2t)$ and $f\left(\frac{t}{2}\right)$.



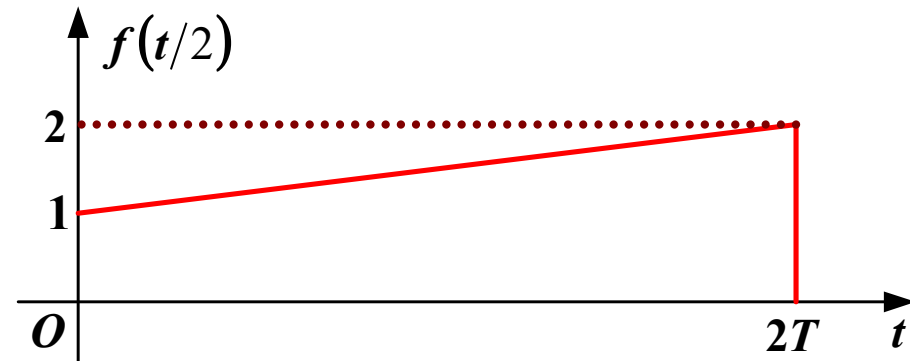
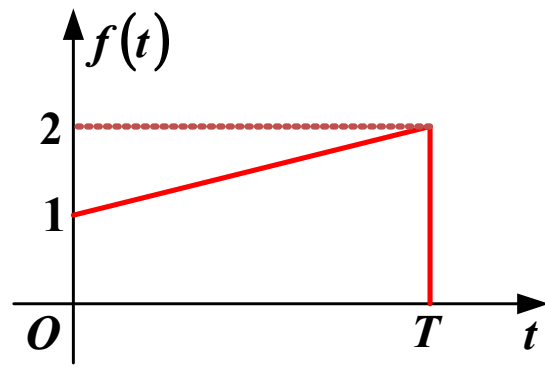
Same argument, same function value				Find new coordinates	
t	$f(t)$	$2t$	$f(2t)$	t	$f(2t)$
0	1	0	1	0	1
T	2	T	2	$T/2$	2

$t \rightarrow 2t$, Time scale increases, waveform compresses

3.2 Operations on continuous-time signals

Example 2

$$f(t) \rightarrow f\left(\frac{t}{2}\right)$$



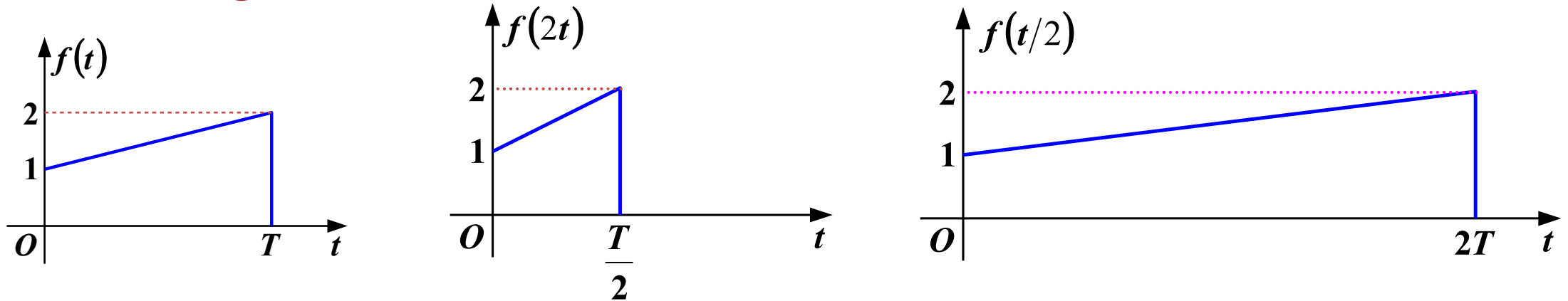
Same argument, same function value					Find new coordinates	
t	$f(t)$	$t/2$	$f(t/2)$		t	$f(t/2)$
0	1	0	1		0	1
T	2	T	2		$2T$	2

$$t \rightarrow \frac{t}{2}$$

Time-scale compression, waveform expansion

3.2 Operations on continuous-time signals

(3) Scaling



$$f(t) \rightarrow f(at) \begin{cases} a > 1 & \text{compression} \\ 0 < a < 1 & \text{expansion} \end{cases}$$

Multiplying the time variable by a coefficient is equivalent to changing the observation time scale

(4) General form of independent variable transformation

$$f(t) \rightarrow f(at \pm b)$$



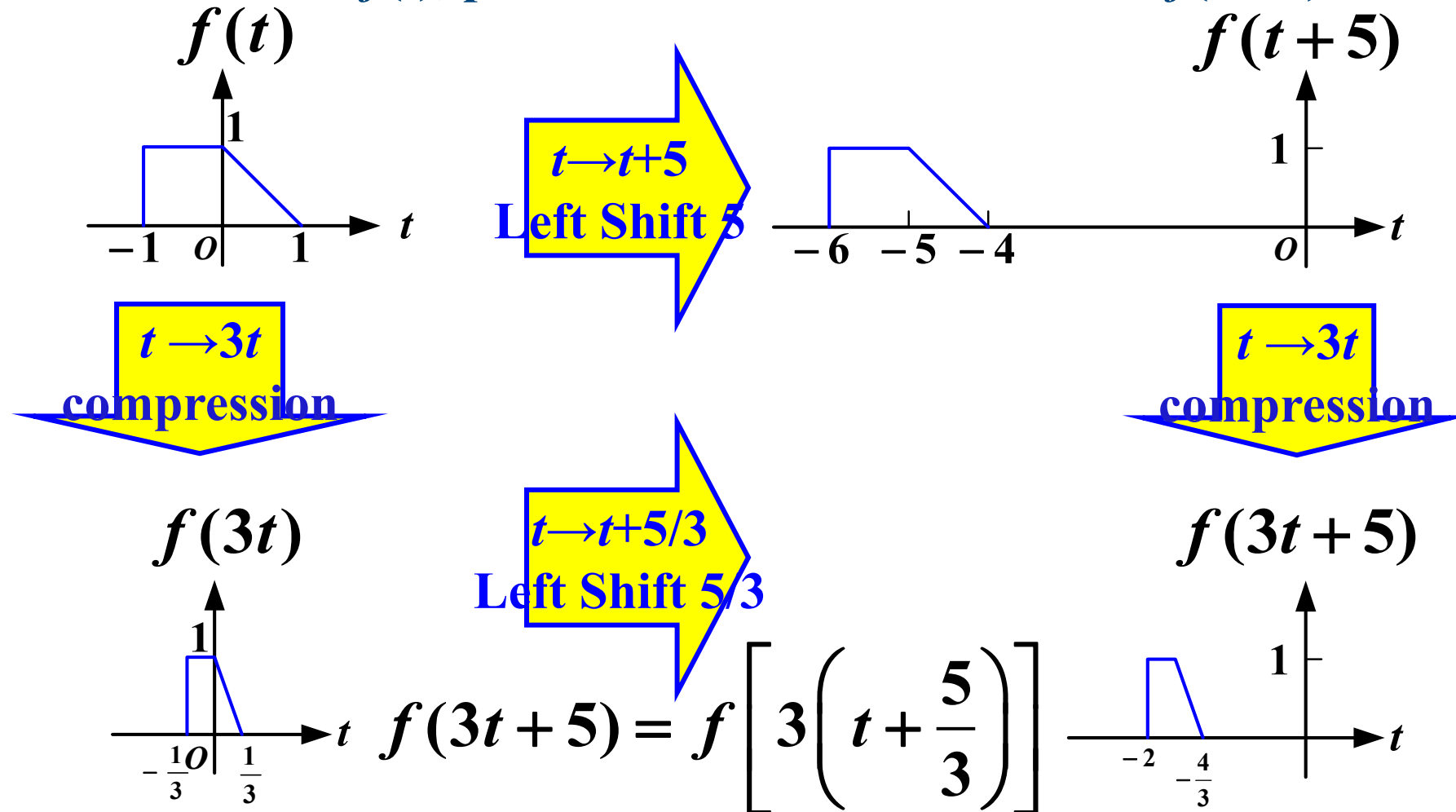
All transformations are applied to t (the independent variable)

3.2 Operations on continuous-time signals



Example 3

Given the waveform of $f(t)$, plot the transformed waveform $f(3t+5)$



Summary

1. Dependent Variable Operations

- Addition and Multiplication
- Differentiation and Integration

2. Independent Variable Transformations

- Time Shifting (Translation)
- Time Reversal
- Time Scaling

$$f(t) \rightarrow f(at \pm b)$$

All transformations are applied to t (the independent variable)